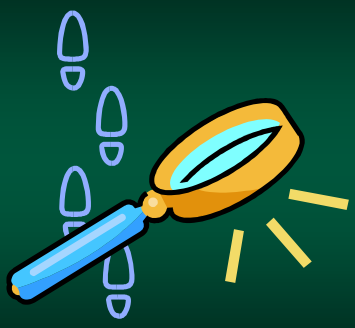
Binary Search & Sorting

Part 8

1



Binary Search

Cutting the problem in half… many times

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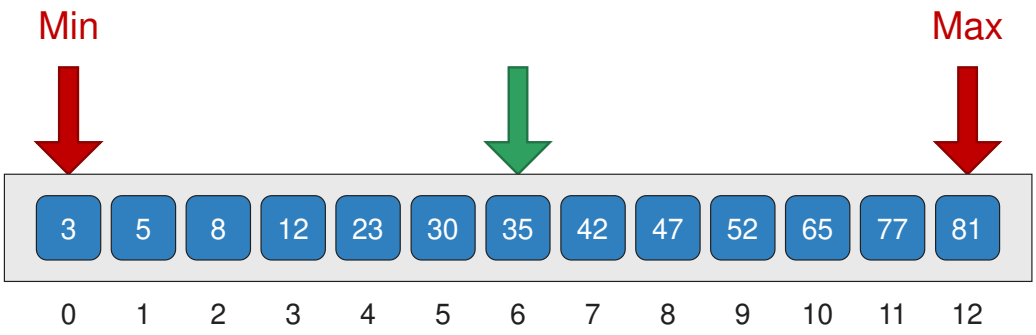
Binary Searching

- A *binary search* is a fast and efficient way to search an array
- Algorithm works like the classic "secret number game"
- Requires that the array is sorted before the search

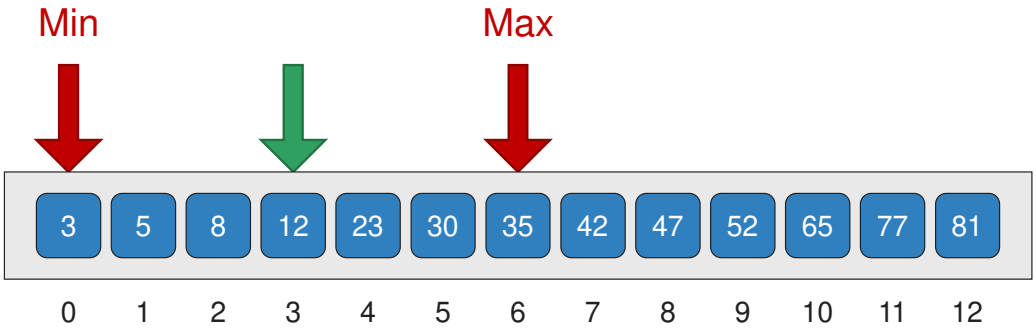
How it Works

- Starts knowing the max & min values
  - in the case of arrays, this is the min and max index
  - in the number game, it is the min and max value
- Algorithm continues
  - it looks at the midpoint between the first and last
  - if the value > target, the max is set to the midpoint
  - if the value < target, the min is set to the midpoint
  - *this eliminates half of the numbers each iteration*

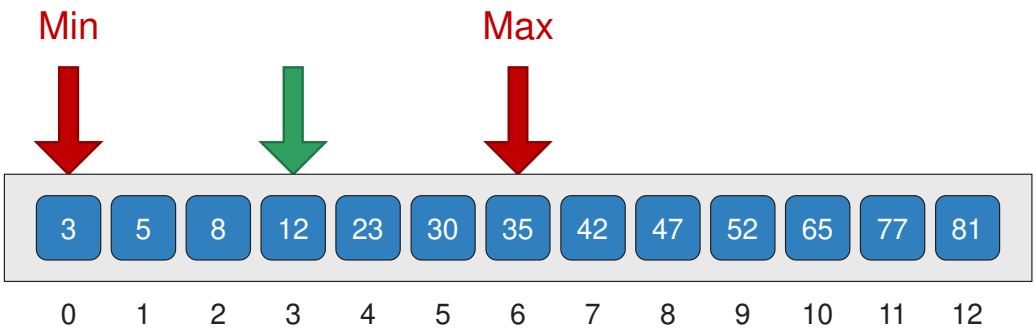
Binary Example: Find 30



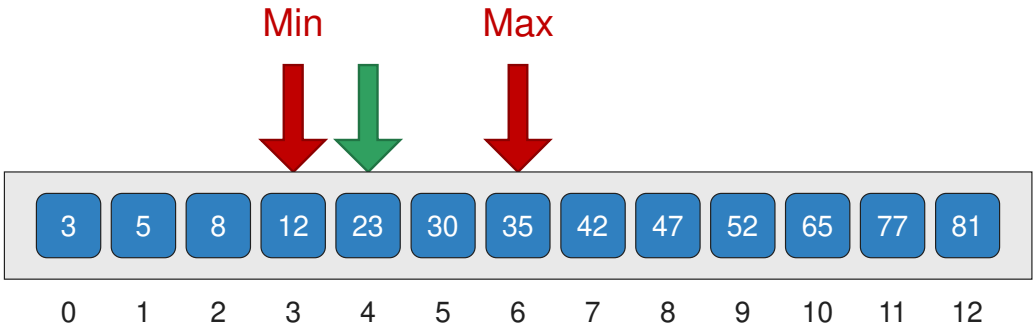
Binary Example: Find 30



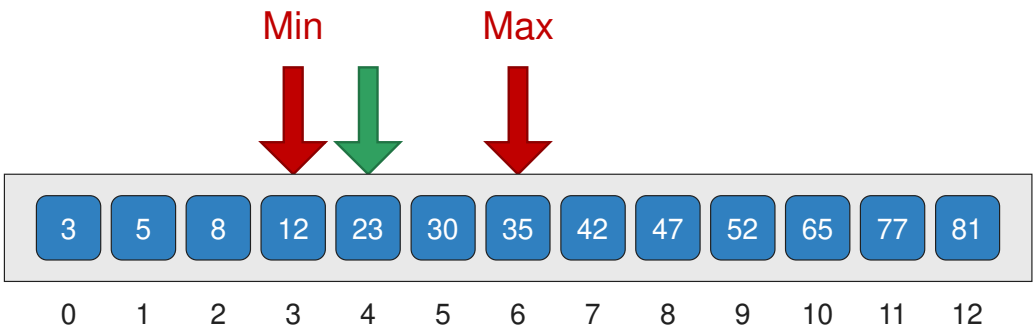
Binary Example: Find 30



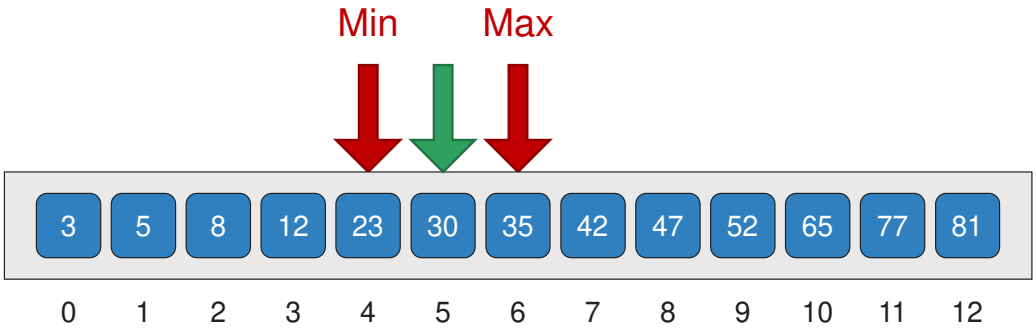
Binary Example: Find 30



Binary Example: Find 30



Binary Example: Find 30

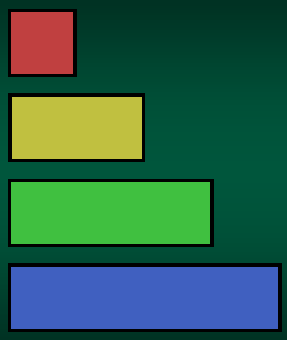


Benefits

- The binary search is incredibly efficient and **absolutely necessary** for large arrays
- Any item can be found only $\log_2(n)$ searches! It is O(log n)
- However, since array must be sorted, sorting algorithms are equally vital

Maximum # of Searches

Array Size	Sequential	Binary
10	10	4
100	100	7
1,000	1,000	10
10,000	10,000	14
100,000	100,000	17
1,000,000	1,000,000	20
10,000,000	10,000,000	24
100,000,000	100,000,000	27
1,000,000,000	1,000,000,000	30



Sorting

Bringing Order of "Chaos"

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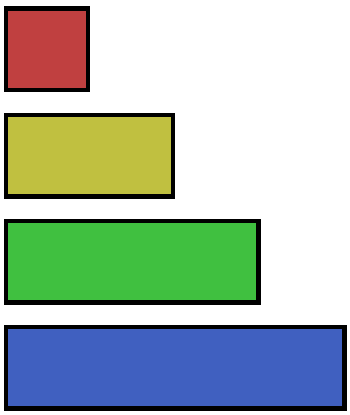
Sorting

- It is useful (and efficient) to *sort* a list of data – to put it in specific order
- There are multiple sorting algorithms which get complex as they become more efficient



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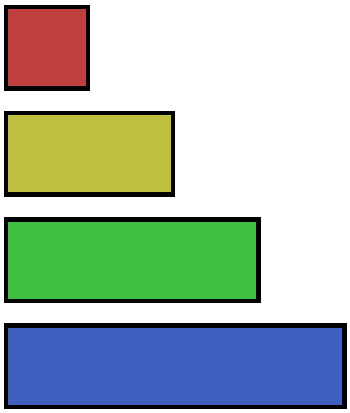
Sorting

- Examples:
  - sorting scores by highest to lowest
  - sorting filenames in alphabetical order
  - sorting students by their student-id

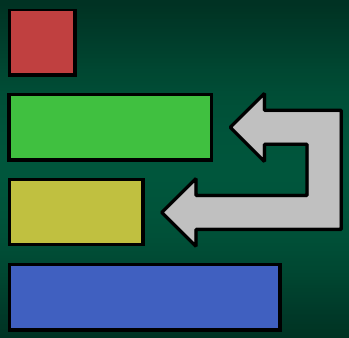


Sorting Algorithm Attributes

1. Time Complexity
   - Big-O classification
   - naturally, the smallest classification is better
2. Auxiliary space
   - how extra much memory is needed to run the algorithm
   - some algorithms require extra memory – *perhaps as large as the array itself*

Sorting Algorithm Attributes

3. Stable
   - what happens when two array elements, a and b, have the same sort value?
   - if a is initially stored before b, a "stable" sort will not change their relative positions
4. Online
   - elements can be added at the same time that the data is being sorted
   - data can be *streamed* into the array at runtime

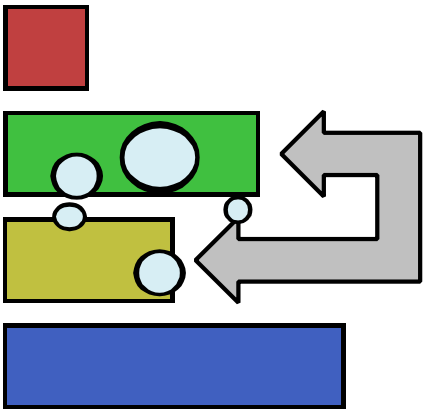


Bubble Sort

Carbonated Sorting

Bubble Sort

- The *bubble sort* is one of the least efficient algorithms …but it is easy to understand
- Basic approach
  - "lighter" elements “bubble up” to the top of the array
  - "heavier" items sink to the bottom



How It Works

- Consists of two For Loops
- Outer loop runs from the first to the last
- Inner loop …
  - runs from the bottom of the array *up* to the top (well, the position of the first loop)
  - it checks every two neighbor elements, if the they are out of order, it swaps them
  - so, the smallest element moves up the array

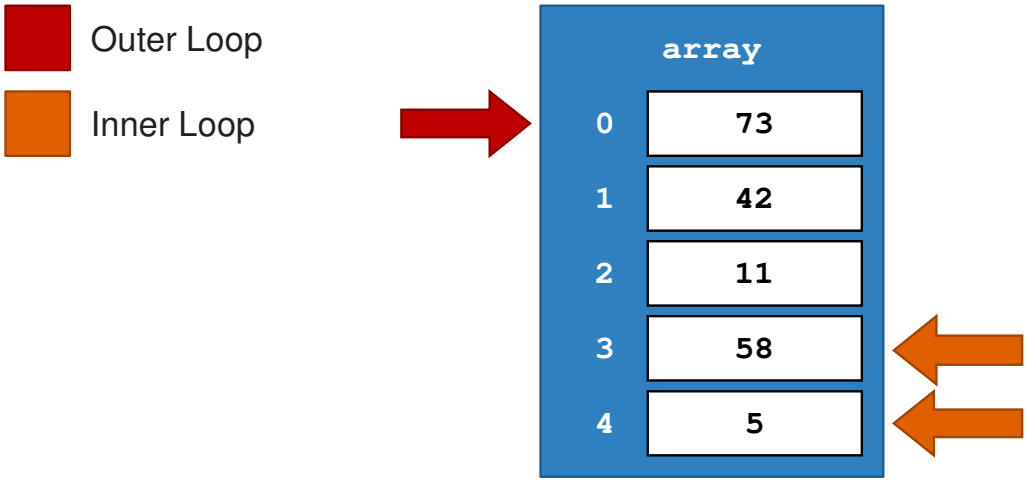
The Bubble Sort

```
for (i = 0; i < array.count - 1; i++)

    for(j = array.count - 1; j > i; j--)
        if array[j - 1] > array[j]
            swap array[j - 1] and array[j]
        end if
    end for

end for
```

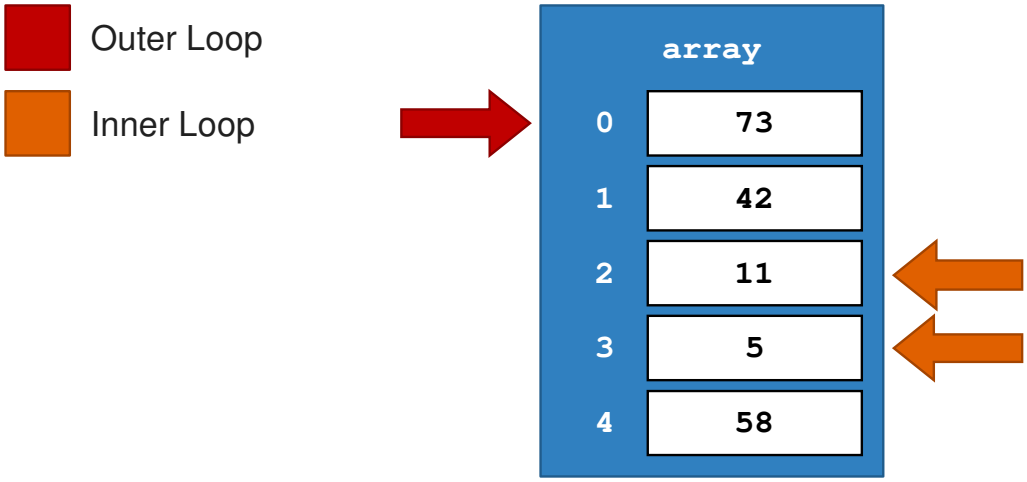

The Bubble Sort (Alternative)

```
for (i = 0; i < array.count - 1; i++)

    for(j = 0; j < array.count - 1 - i; j++)
        if array[j] > array[j + 1]
            swap array[j] and array[j + 1]
        end if
    end for

end for
```

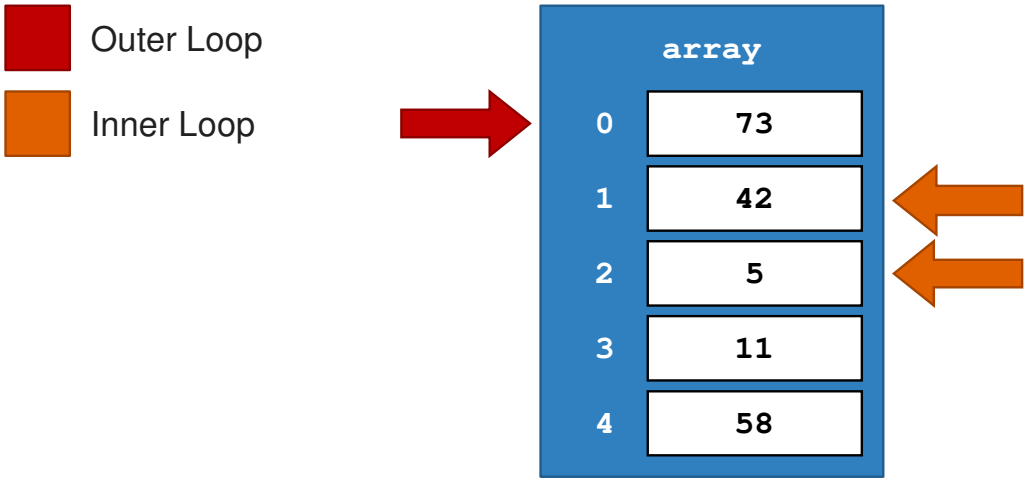
Bubble Sort Example



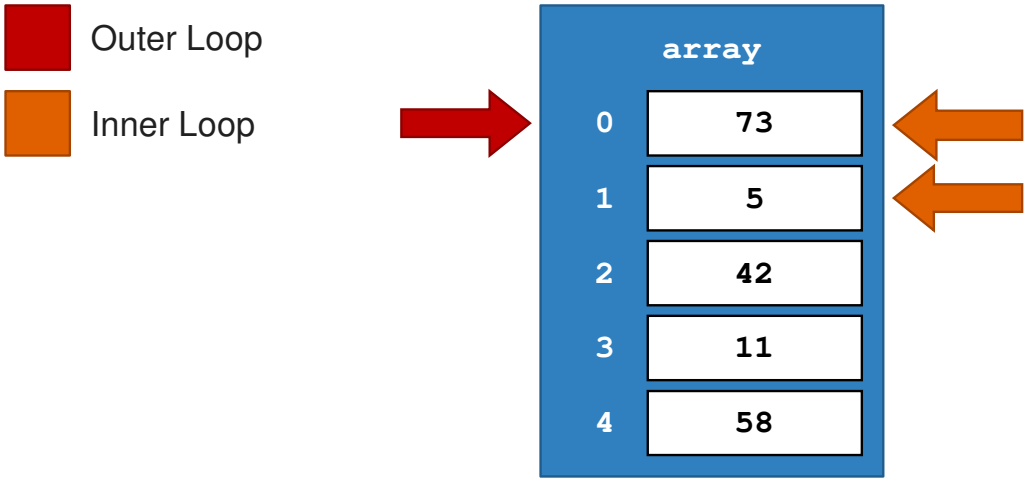
Bubble Sort Example



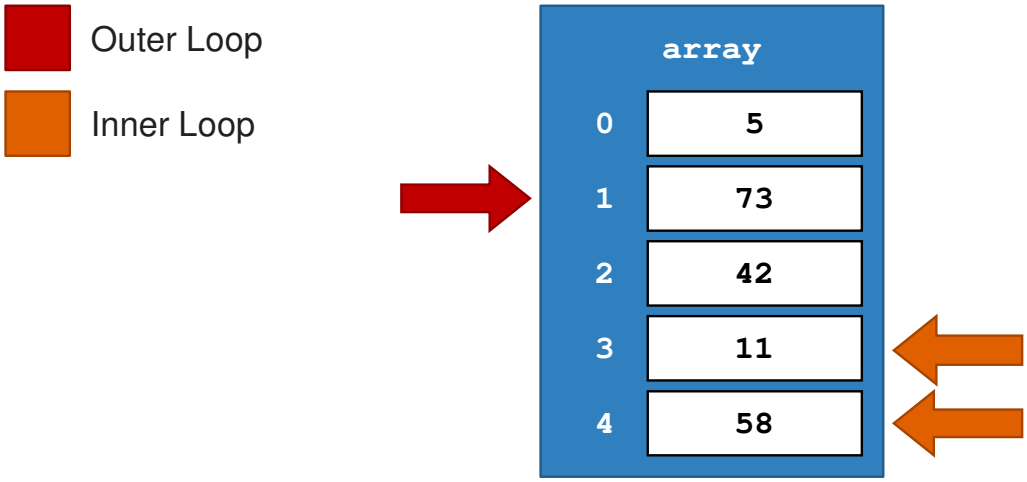
Bubble Sort Example



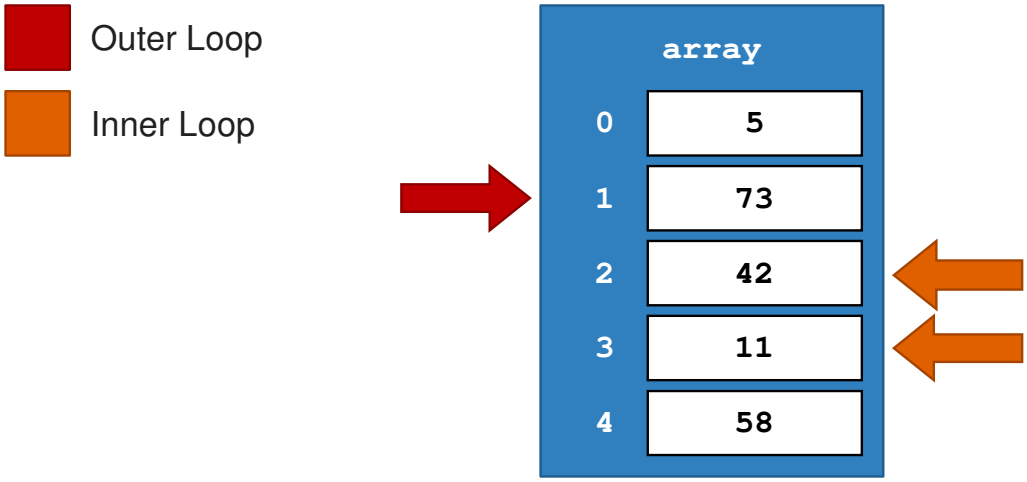
Bubble Sort Example



Bubble Sort Example



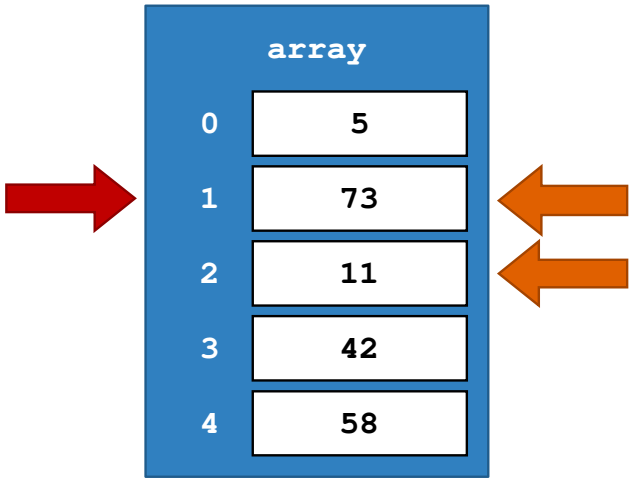
Bubble Sort Example



Bubble Sort Example

Outer Loop

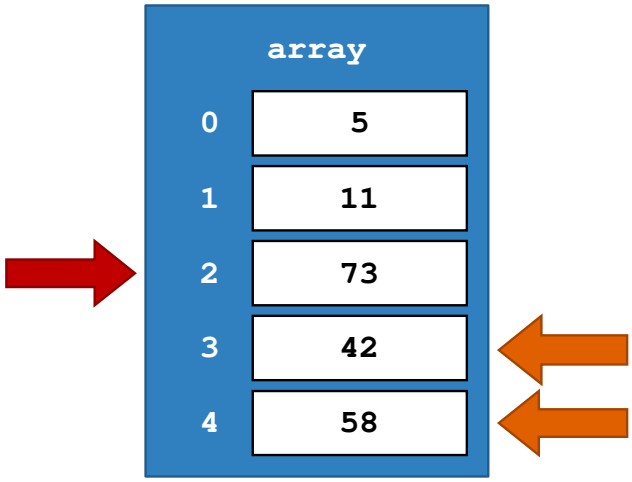
Inner Loop



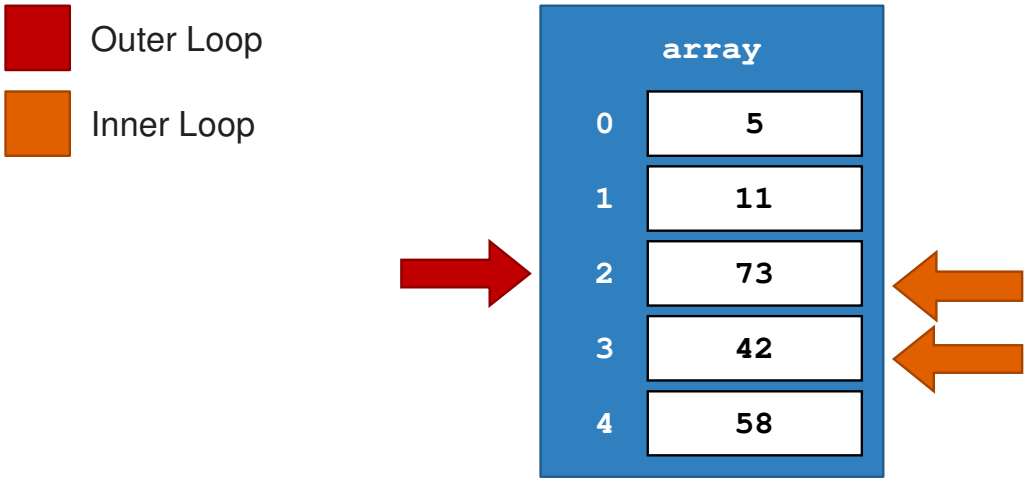
Bubble Sort Example

Outer Loop

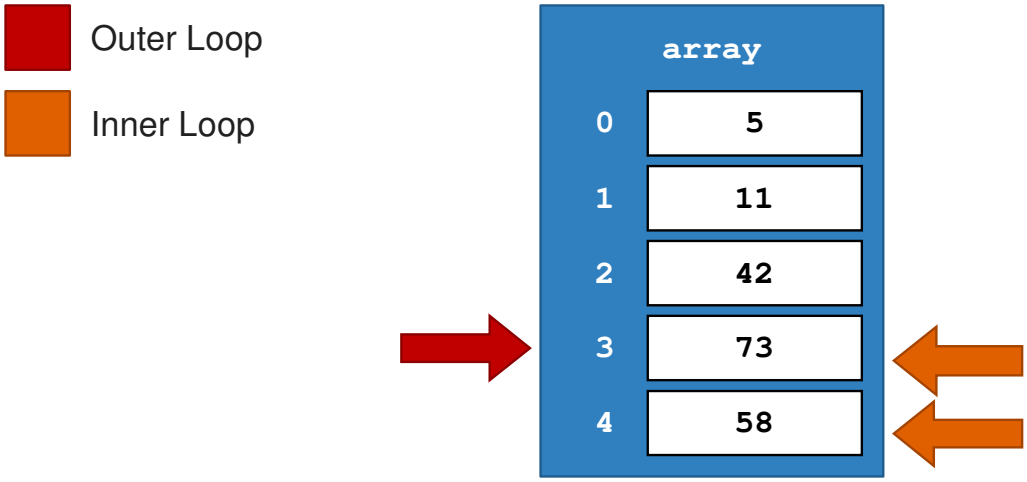
Inner Loop



Bubble Sort Example



Bubble Sort Example



Bubble Sort Example

Outer Loop

Inner Loop

array	
0	5
1	11
2	42
3	58
4	73

Efficiency of the Bubble Sort

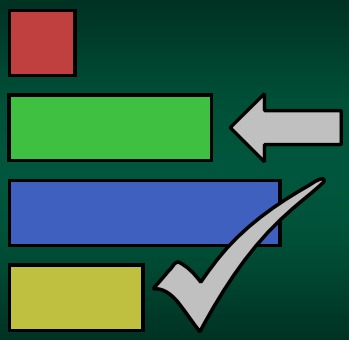
- The Bubble Sort is **extremely** inefficient and only good for tiny arrays
- Since Bubble Sort uses two embedded loops
  - the outer loop looks at all *n* items
  - the inner loop looks at basically *n* items
  - the resulting algorithm gets incredibly less efficient as *n* increases

Efficiency of the Bubble Sort

- The Bubble Sort **$O(n^2)$**
- … two embedded loops that are based on *n*
- … and all that swapping doesn't help either!

Bubble Sort Summary

Bubble Sort	
Time Average	$O(n^2)$
Time Best	$O(n^2)$
Time Worst	$O(n^2)$
Auxiliary space	O(1)
Stable	Yes – Equal element order preserved
Online?	No – Entire array in use



Selection Sort

The Human Way

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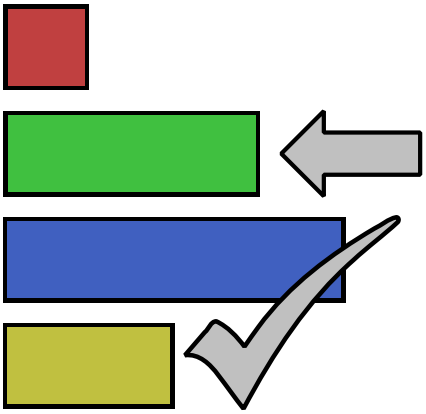
Selection Sort

- The *Selection Sort* is a similar to the Bubble Sort
- However…
  - rather than "bubble up" smaller items, it scans the entire array
  - it finds the smallest element
  - only *then* does it swap the values



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Selection Sort

- Like the Bubble Sort, it consists of two For Loops – one outer and one inner
- Outer loop runs from the first to the last
- Inner loop …
  - starts at the position of the outer loop
  - scans down and finds the *smallest* value
- Then, after the scan, do a single swap

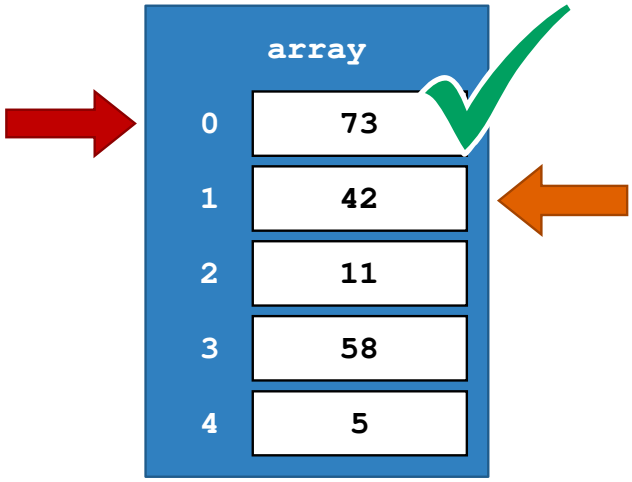
The Selection Sort

```
for (i = 0; i < array.count - 1; i++)
    best = i

    for (j = i; j < array.count; j++)
        if array[j] < array[best]
            best = j
        end if
    end for

    swap array[j] and array[best]
end for
```

Selection Sort Example

Outer Loop

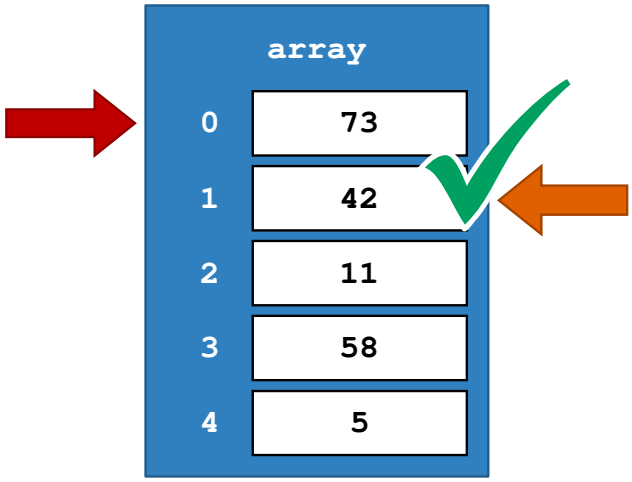
Inner Loop



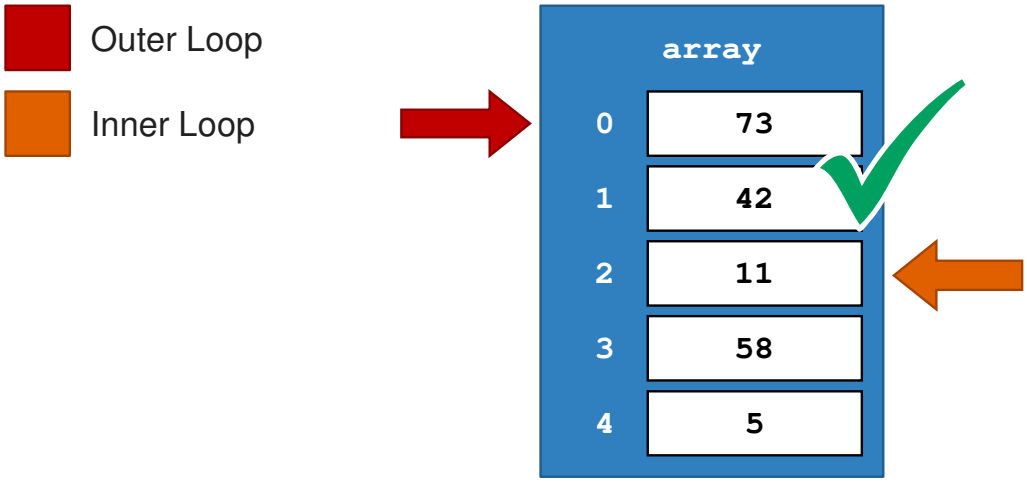
Selection Sort Example: New Best

Outer Loop

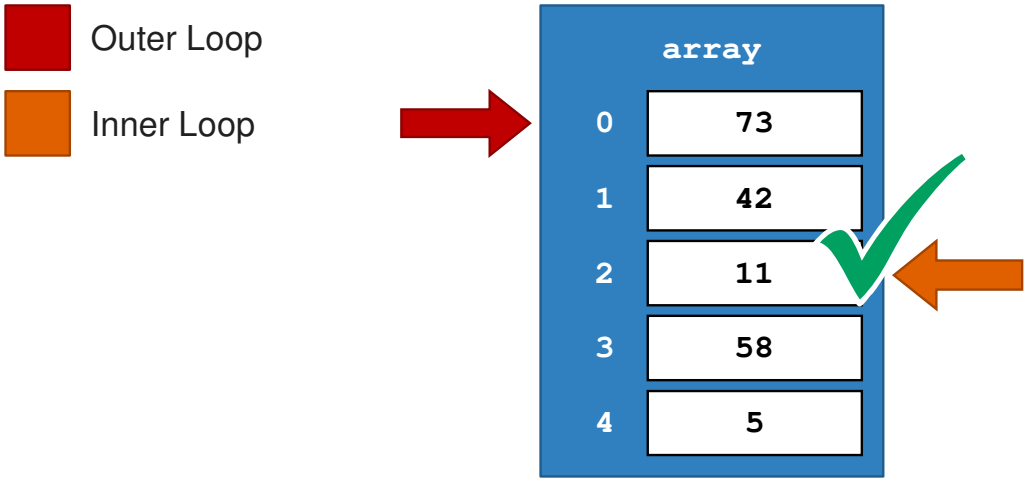
Inner Loop



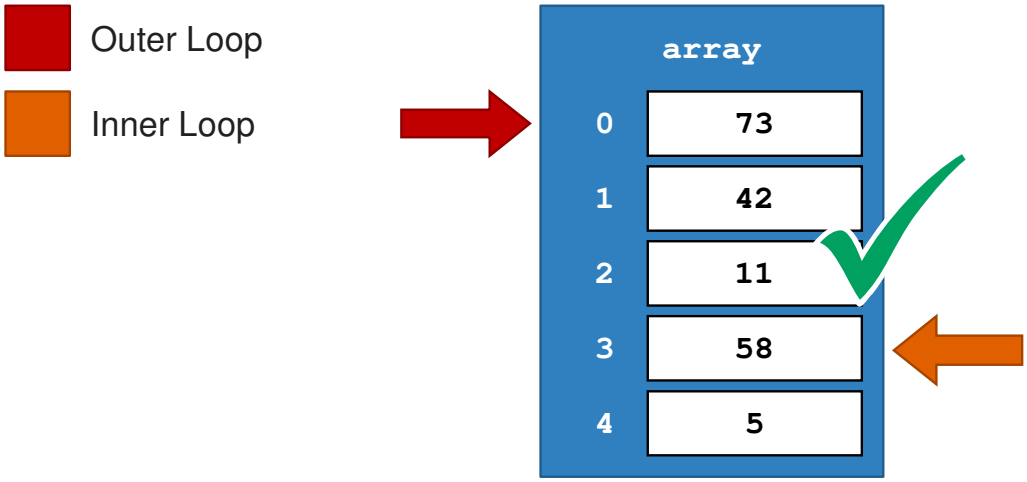
Selection Sort Example



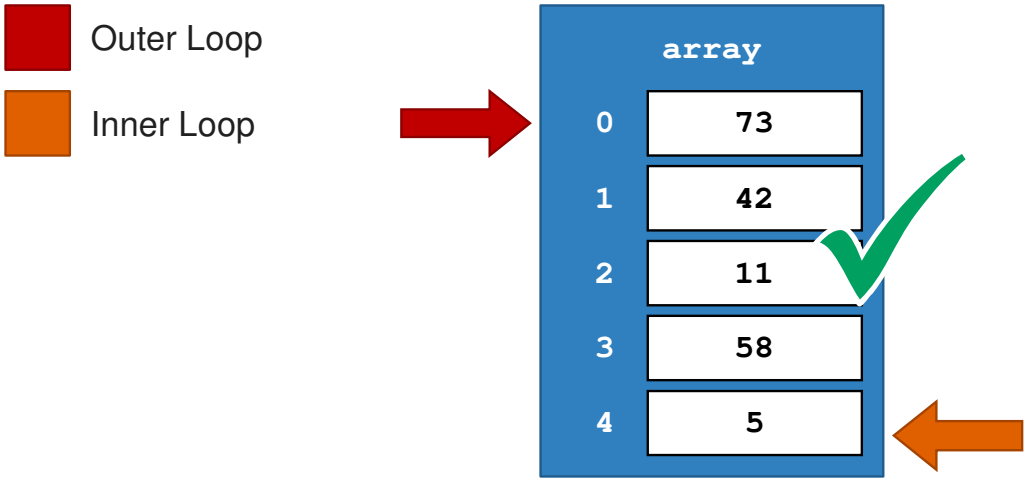
Selection Sort Example: New Best



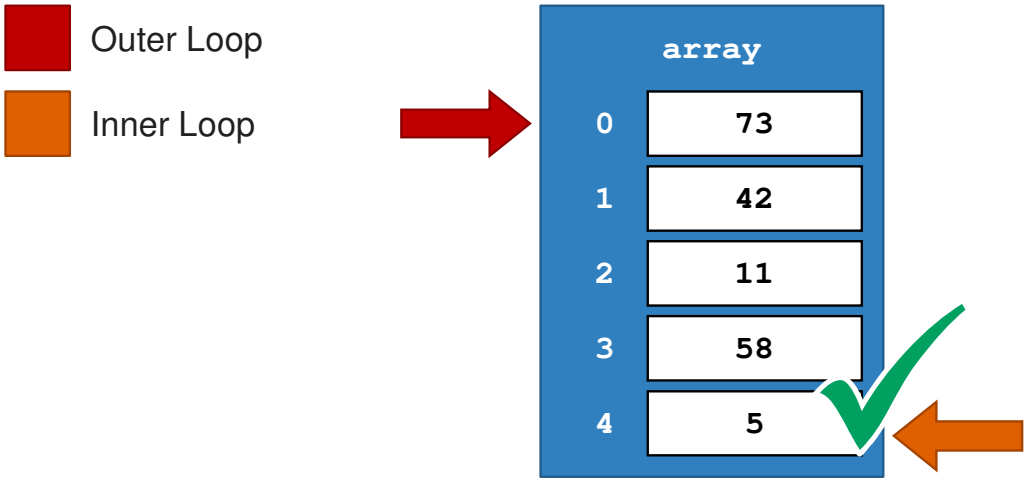
Selection Sort Example



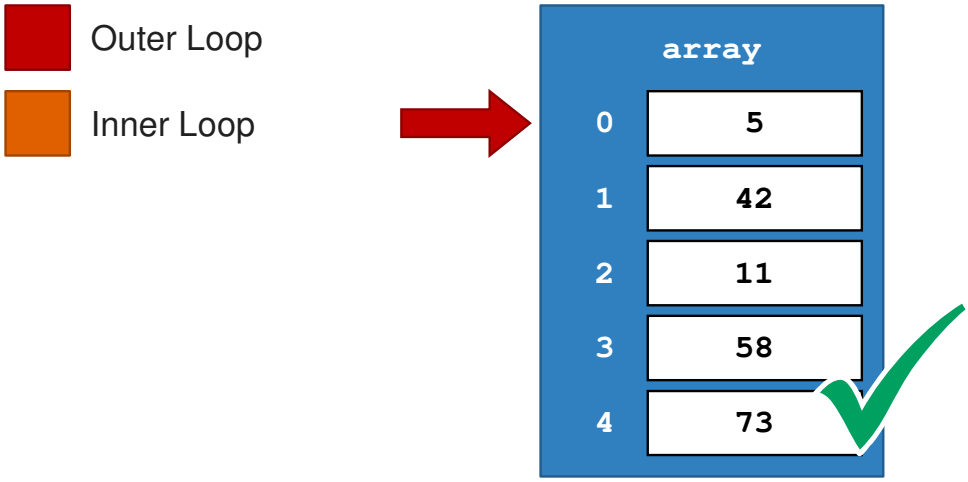
Selection Sort Example



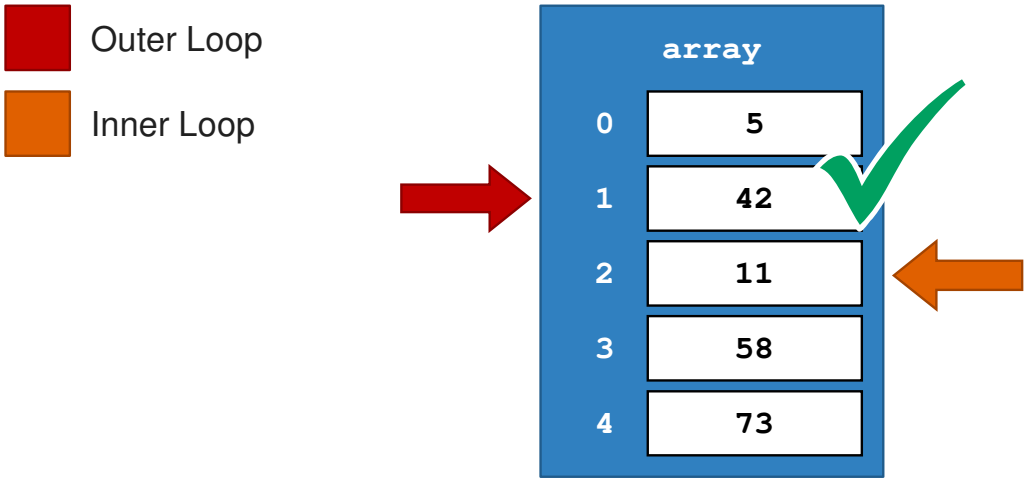
Selection Sort Example: New Best



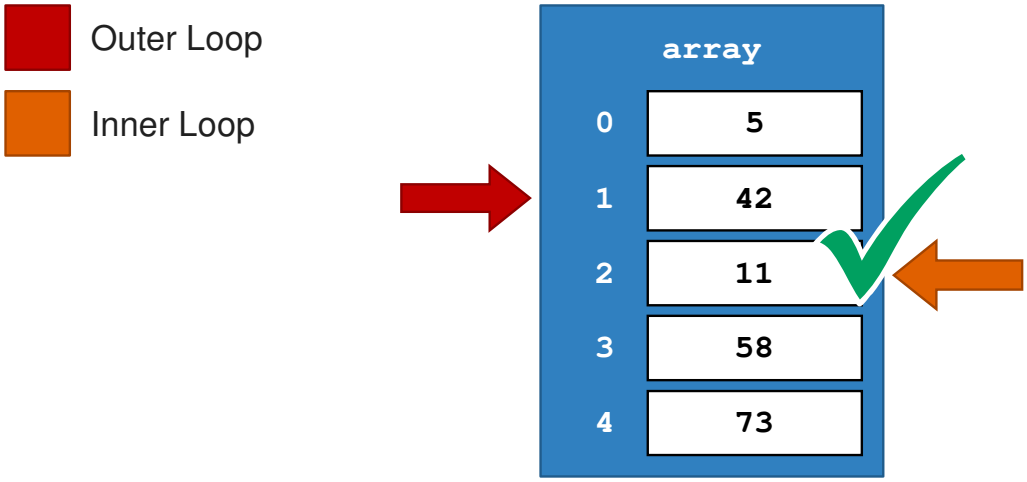
Selection Sort Example: Swapped



Selection Sort Example: Search Again



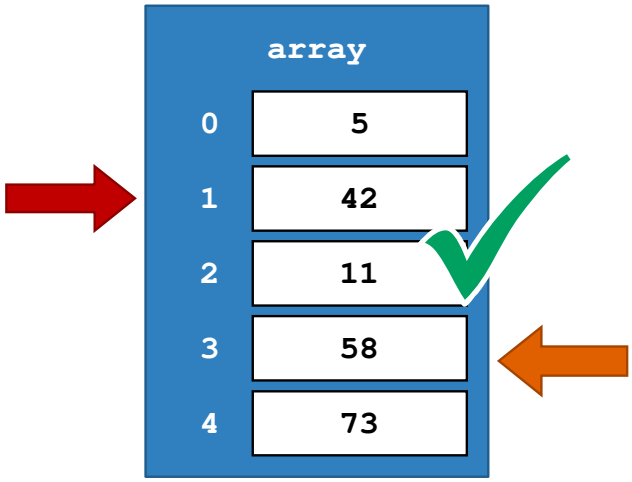
Selection Sort Example: New Best



Selection Sort Example

Outer Loop

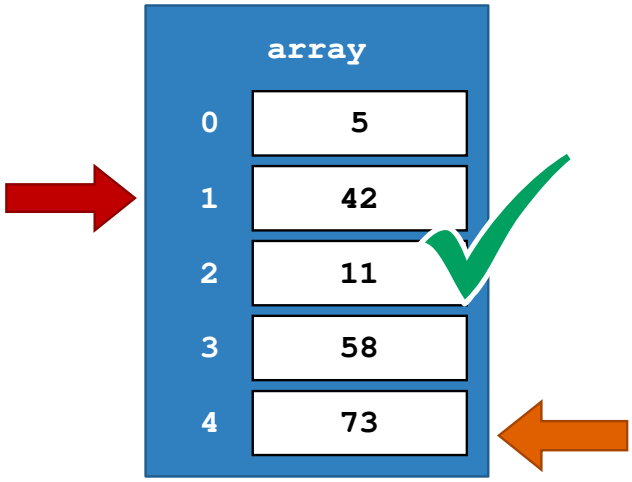
Inner Loop



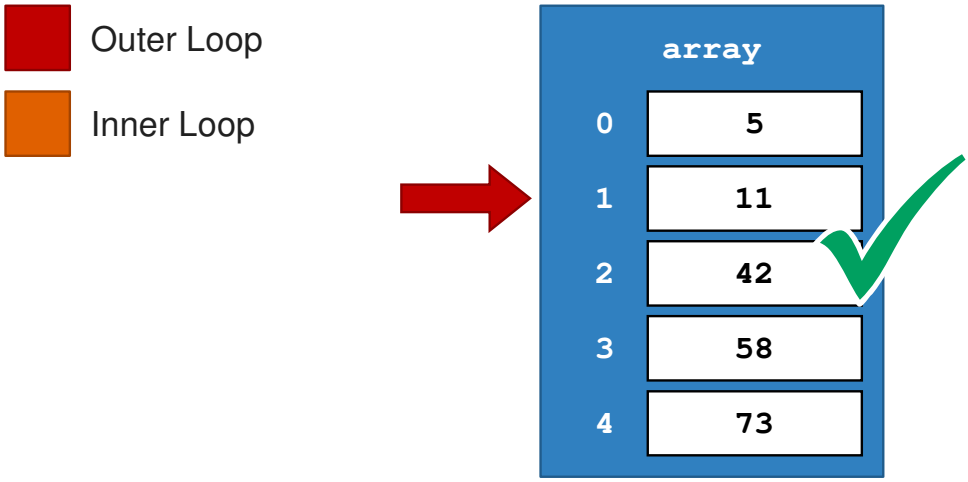
Selection Sort Example

Outer Loop

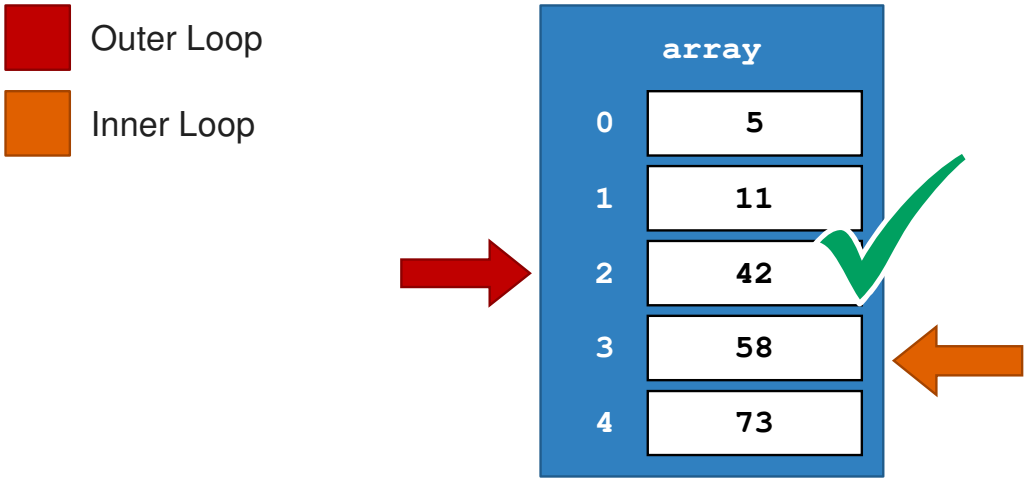
Inner Loop



Selection Sort Example: Swapped



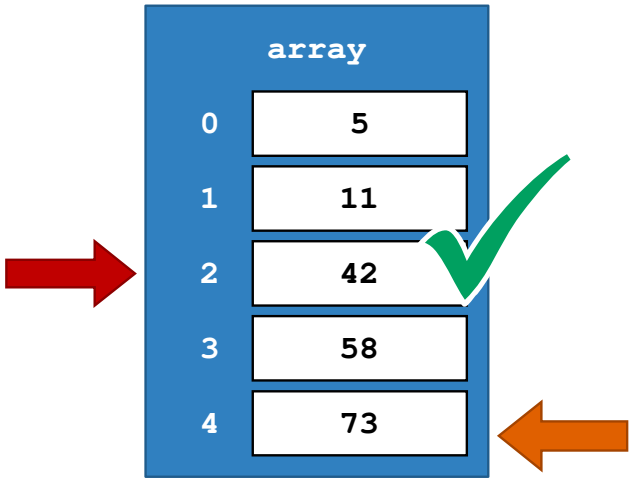
Selection Sort Example: Search Again



Selection Sort Example

Outer Loop

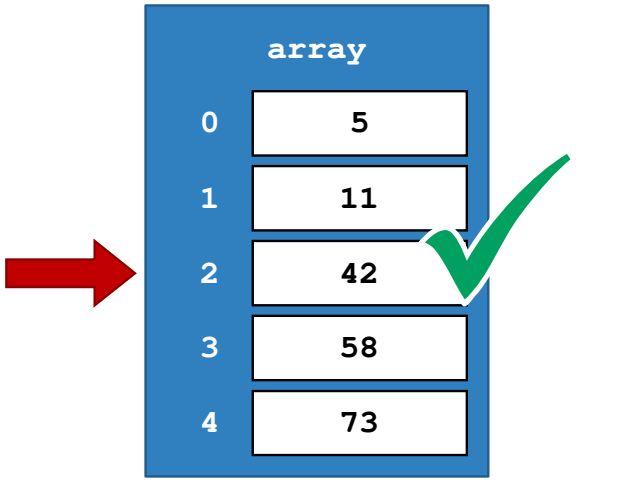
Inner Loop



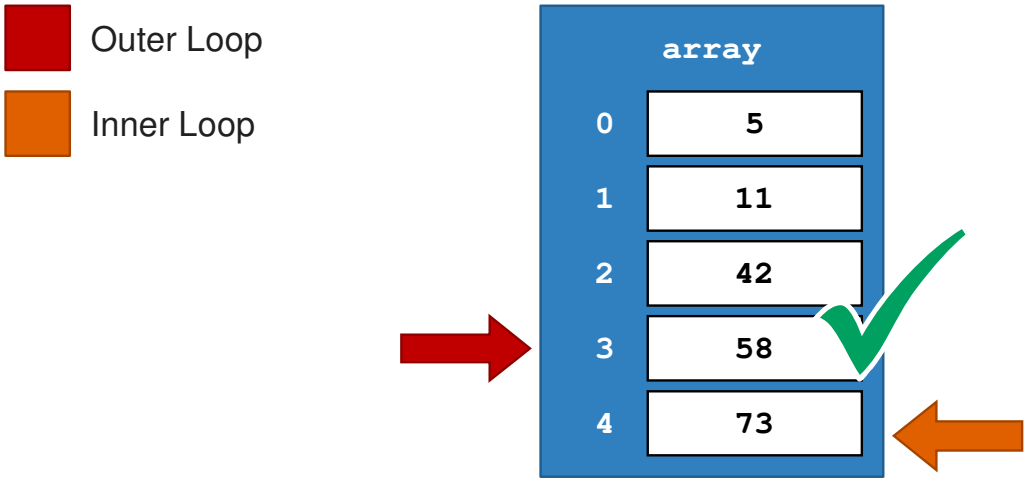
Selection Sort Example: No Swap

Outer Loop

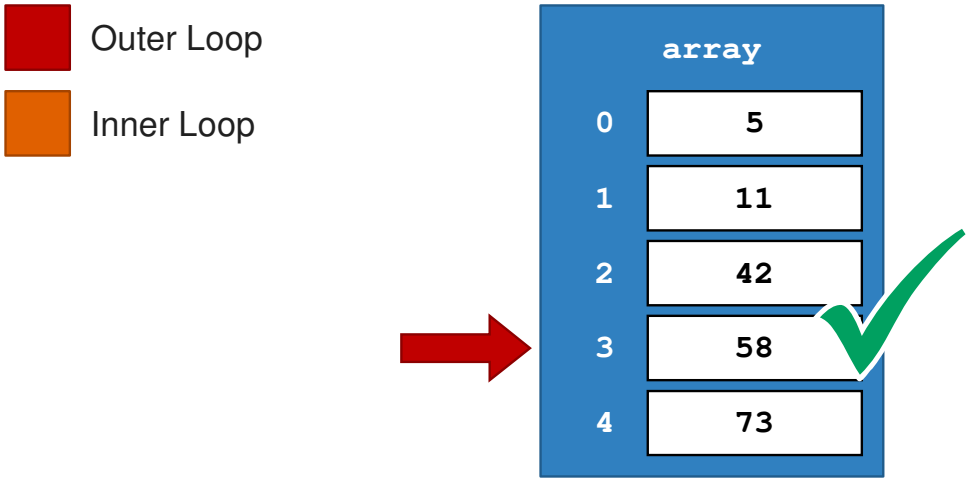
Inner Loop



Selection Sort Example: Search Again



Selection Sort Example: No Swap



Selection Sort Example: Done

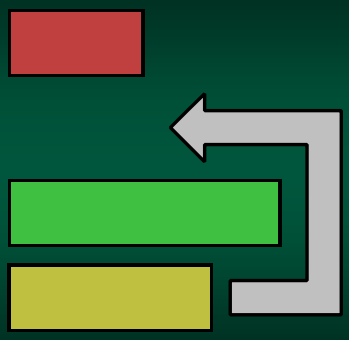
Outer Loop

Inner Loop

array	
0	5
1	11
2	42
3	58
4	73

Selection Sort Summary

Selection Sort	
Time Average	$O(n^2)$
Time Best	$O(n^2)$
Time Worst	$O(n^2)$
Auxiliary space	O(1)
Stable	Yes – Equal element order preserved
Online?	No – Entire array in use



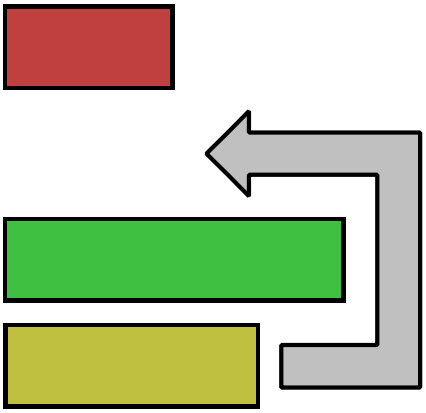
Insertion Sort

Building a sorted array... bit by bit

(er... byte by byte?)

Insertion Sort

- The *Insertion Sort* is a $O(n^2)$ sorting algorithm with several advantages over bubble-sort and selection-sort
- While it is still $O(n^2)$ is far more efficient than the other two



Deck of Cards

- Often, it is compared to sorting a deck of cards
- This is how you would manually sort a row of cards
  - if you start sorting on the left side, you will find a card, move it, and shift the rest of the cards right
  - you build a sorted list a bit at a time – on the left side of your row

How it Works

- The algorithm consists of two loops – one embedded within the other
- The outer loop starts and the top of the array and moves down
- The algorithm builds a sorted array above the outer loop.

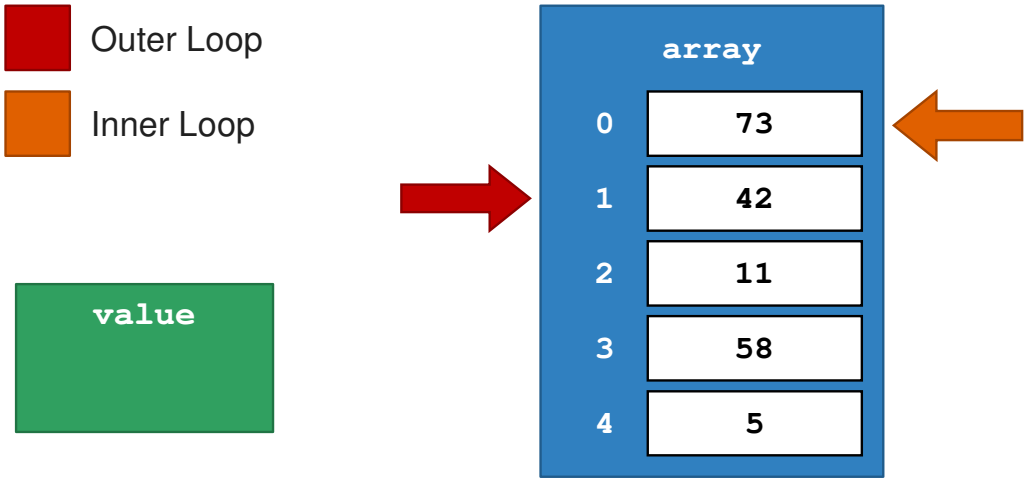
How it Works

- Current array value is saved into a temporary variable
- Inner loop then searches all the values that come before it in the array
- If the value, being looked at, is larger than the saved value, it's moved down

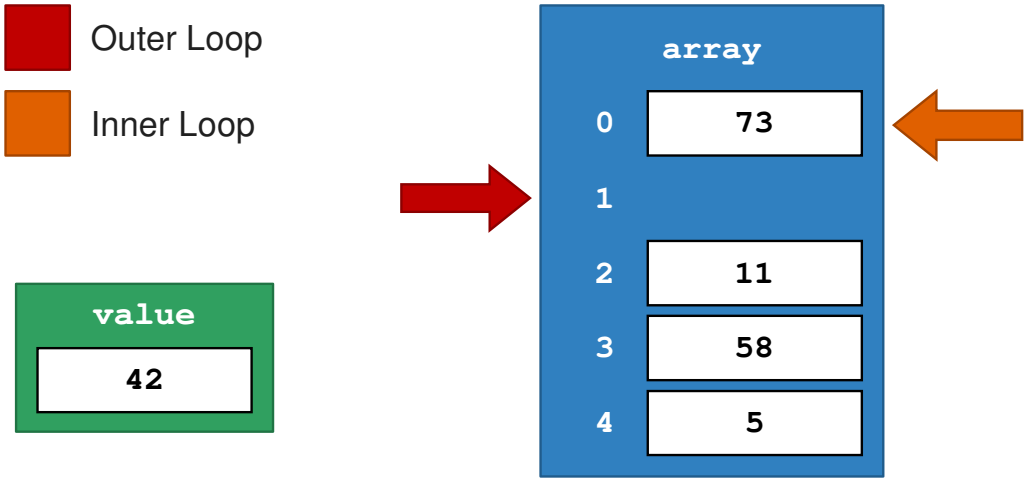
The Insertion Sort

```
for (i = 1; i < array.count; i++)
   value = array[i]

   j = i - 1
   while  j >= 0 and array[j] > value
      array[j + 1] = array[j]
      j--
   end while

   array[j + 1] = value
end for
```

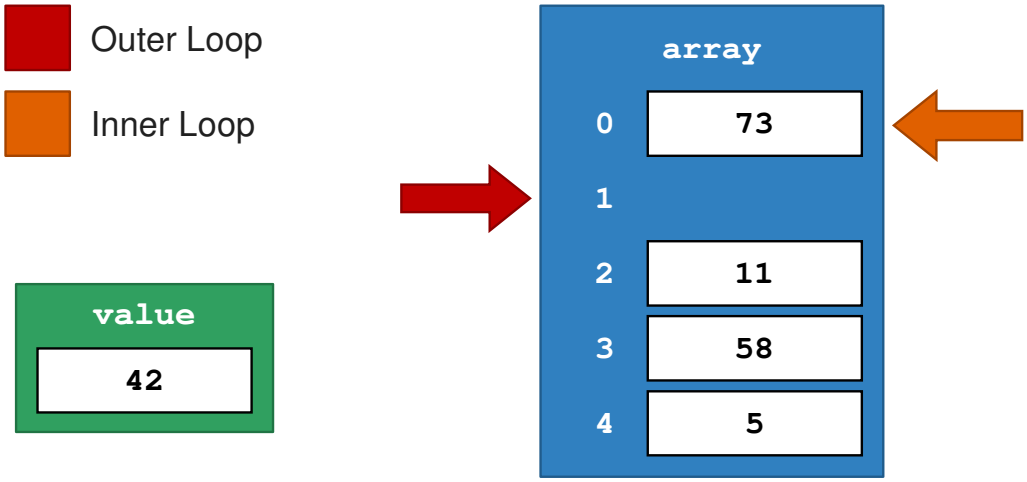
Insertion Sort Example



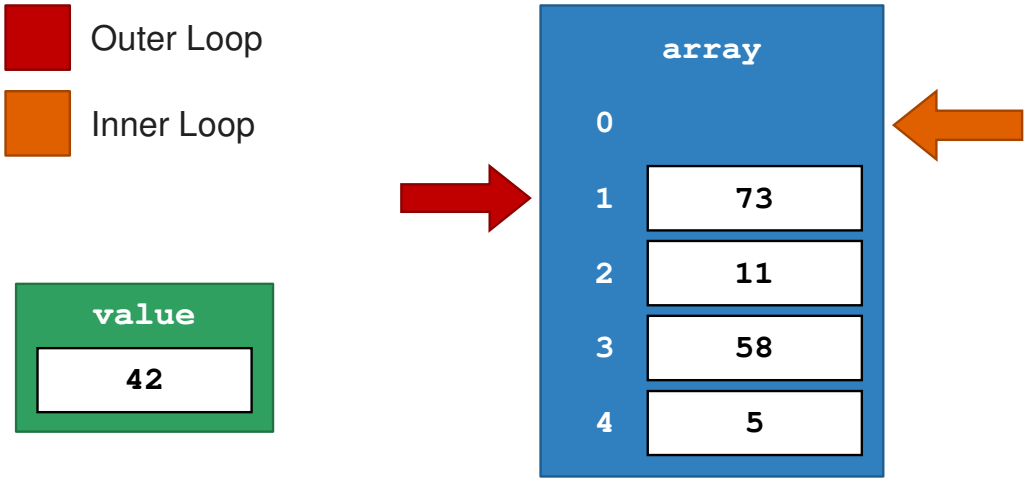
Insertion Sort Example



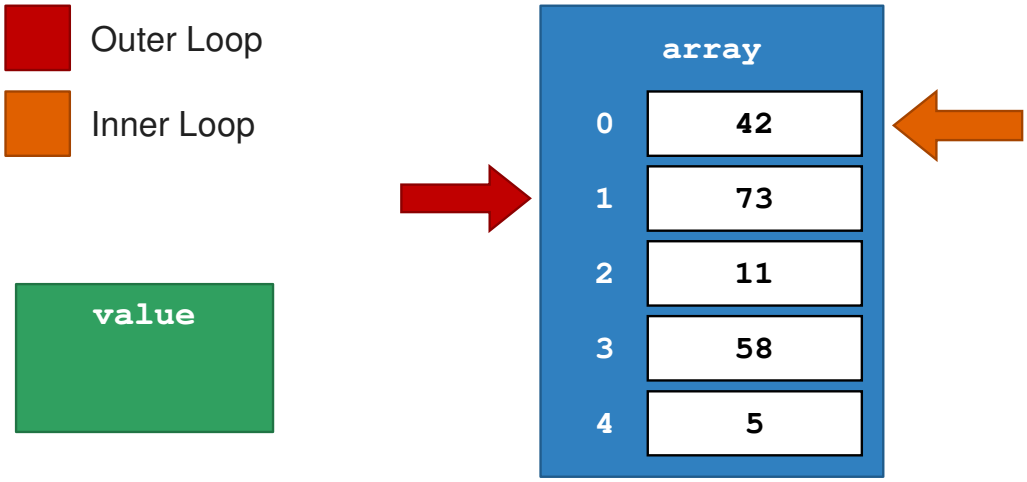
Insertion Sort Example



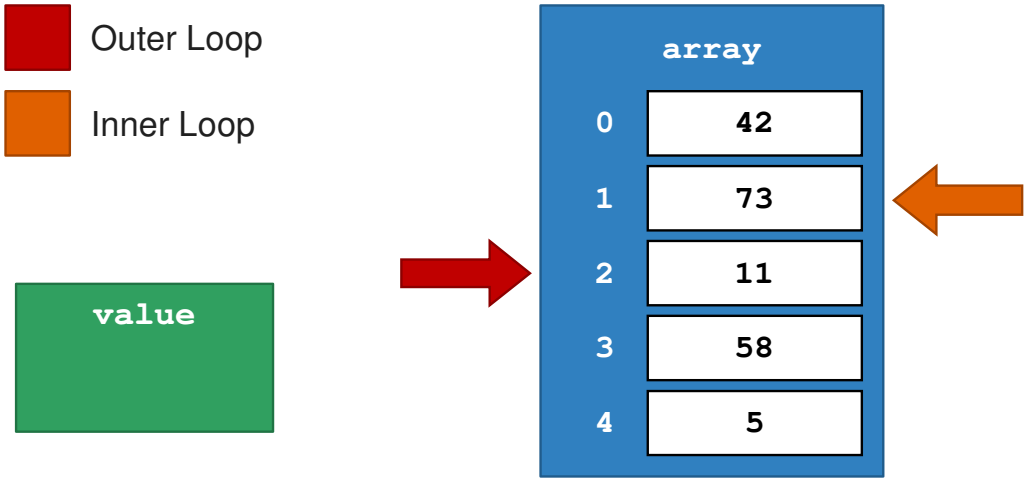
Insertion Sort Example: Return "Card"



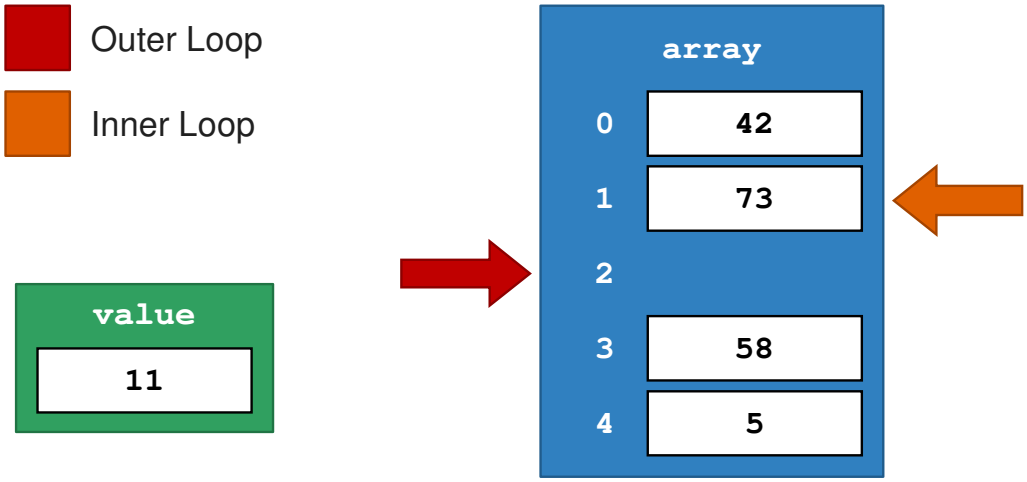
Insertion Sort Example: Return "Card"



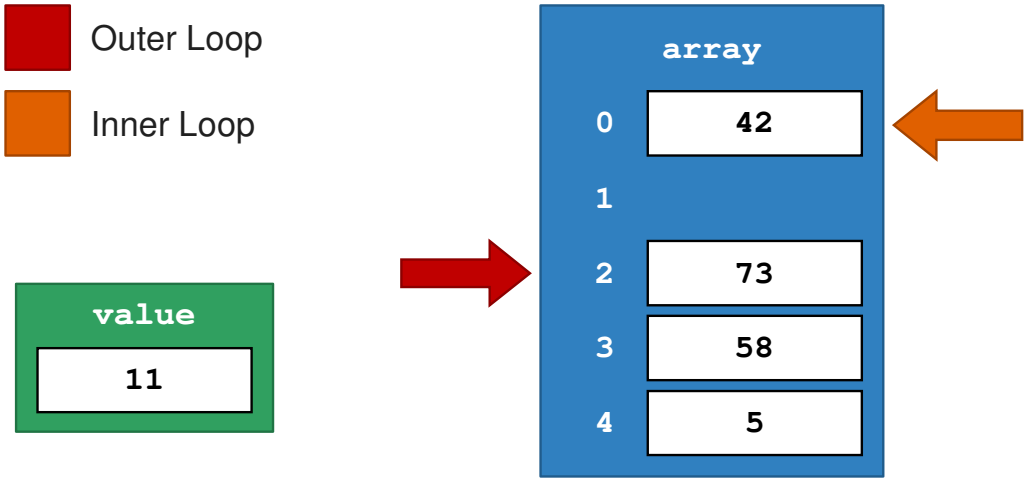
Insertion Sort Example: Next Pass



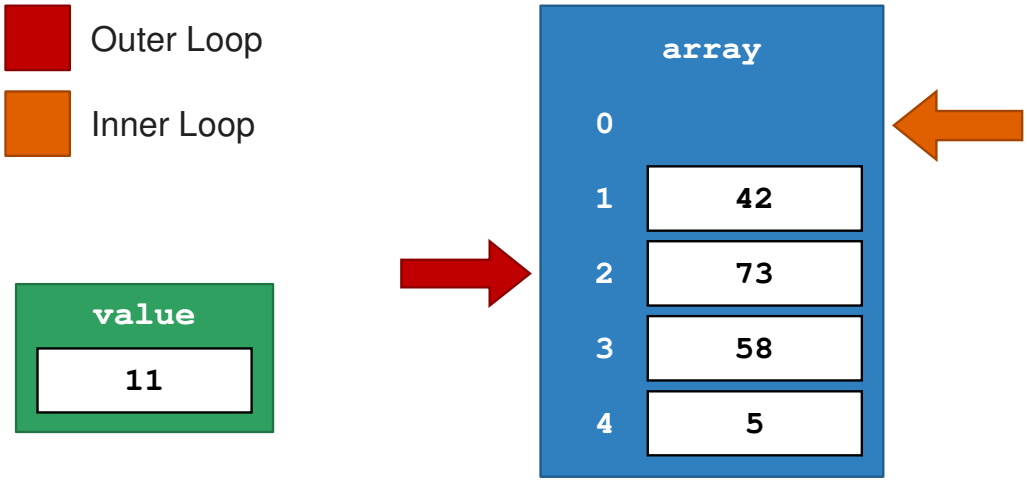
Insertion Sort Example



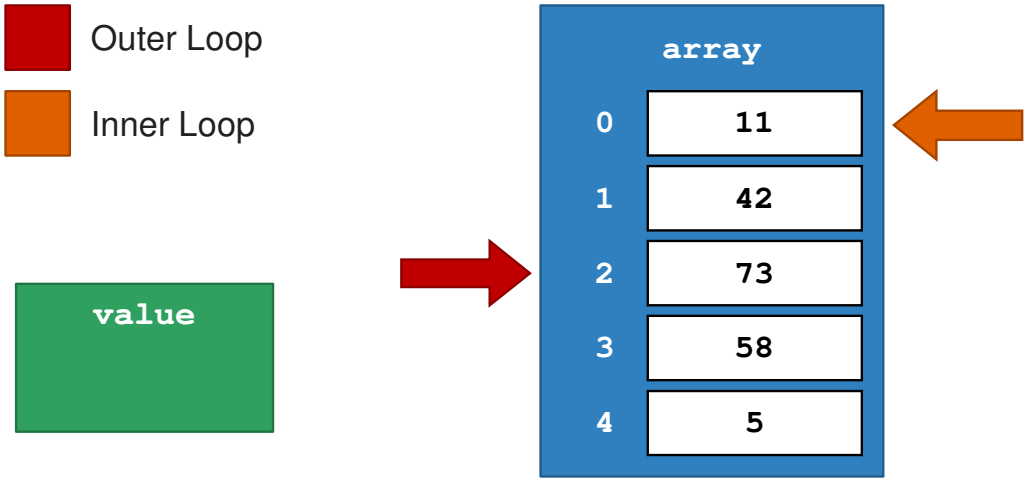
Insertion Sort Example



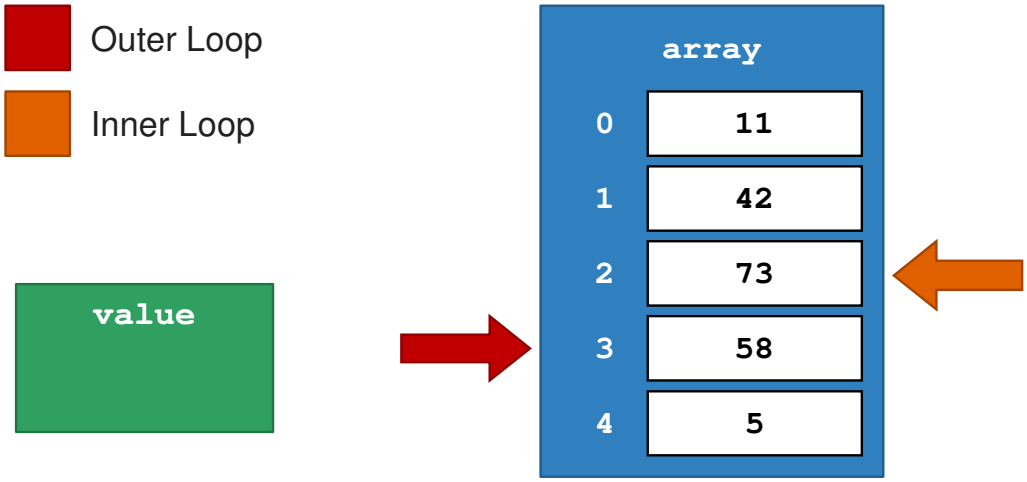
Insertion Sort Example



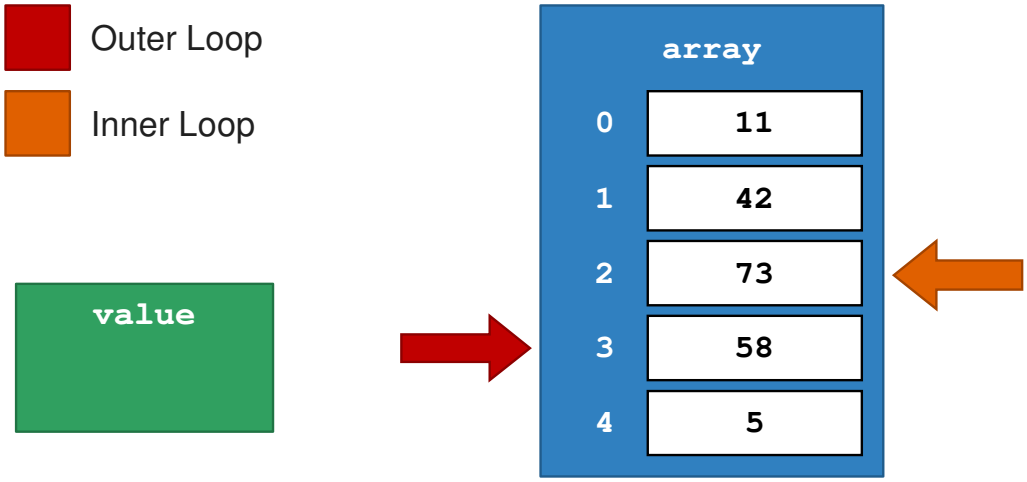
Insertion Sort Example: Return "Card"



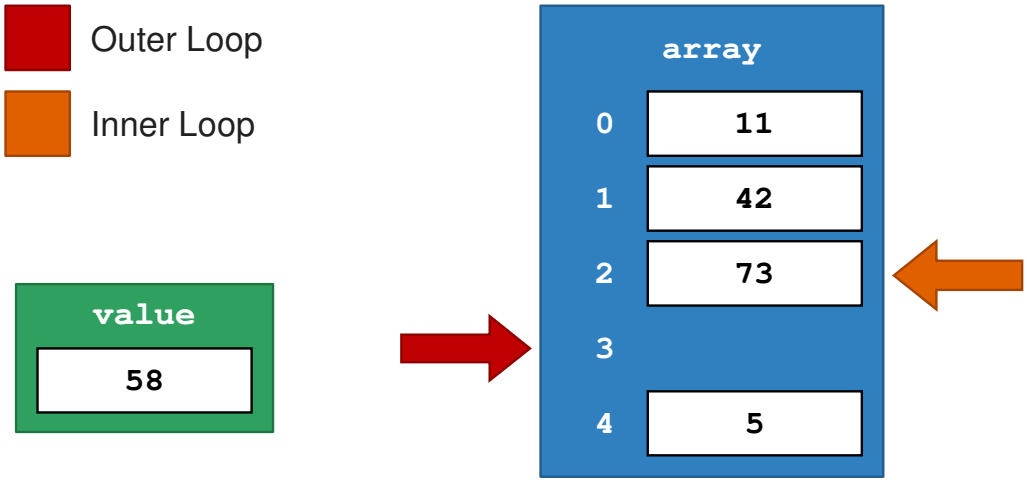
Insertion Sort Example: Next Pass



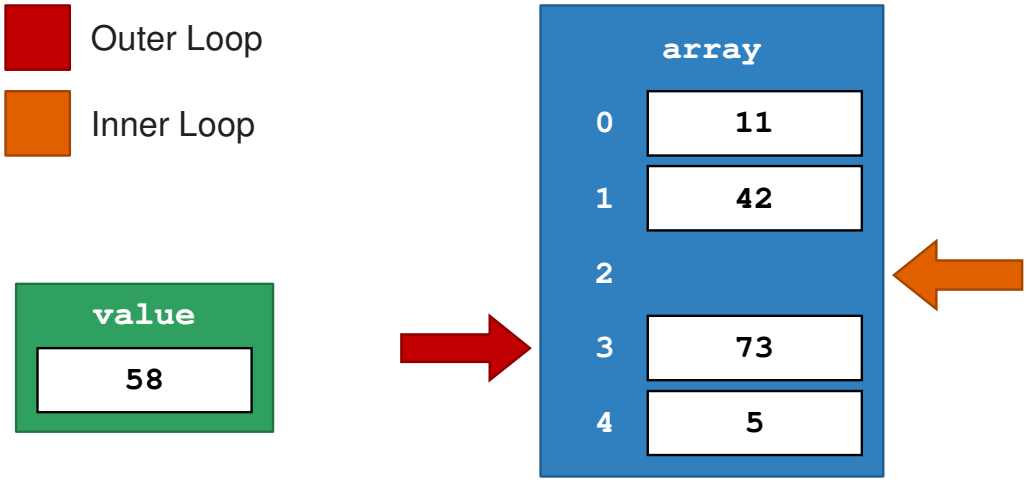
Insertion Sort Example: Next Pass



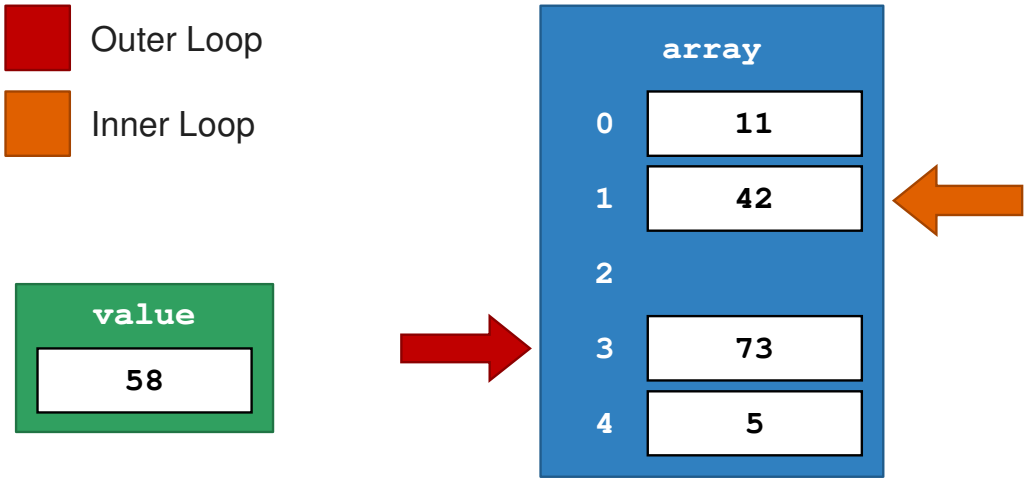
Insertion Sort Example



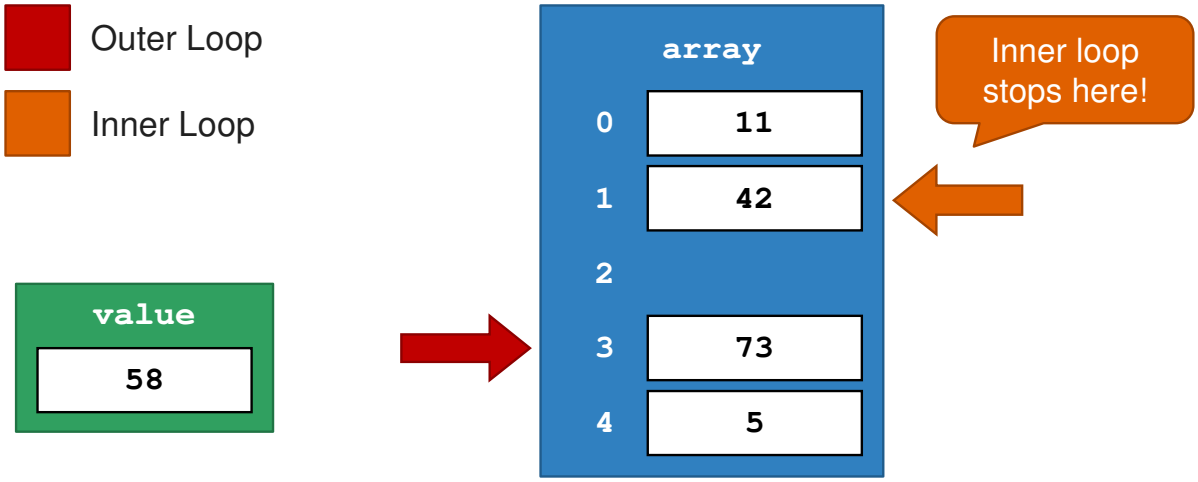
Insertion Sort Example



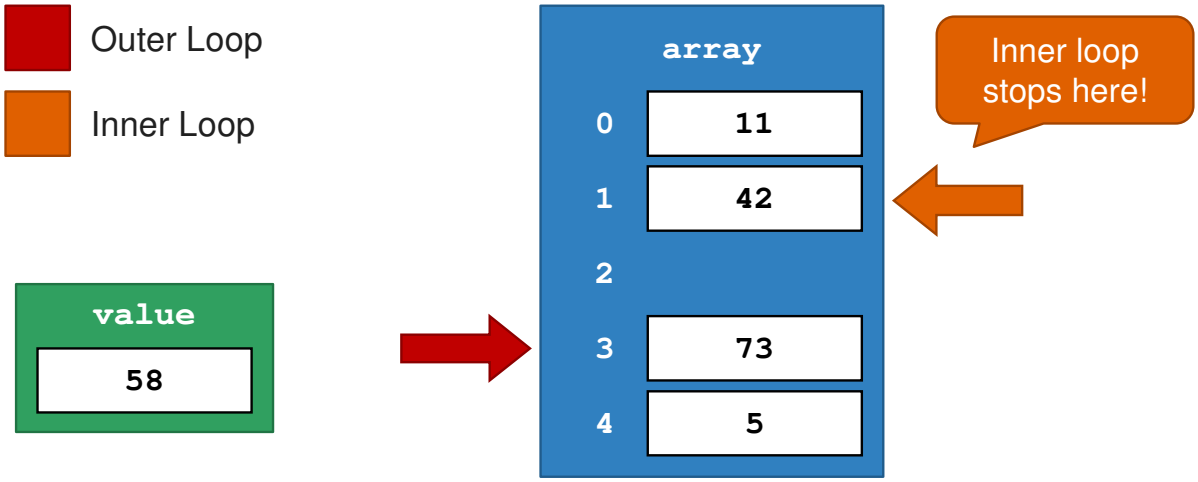
Insertion Sort Example



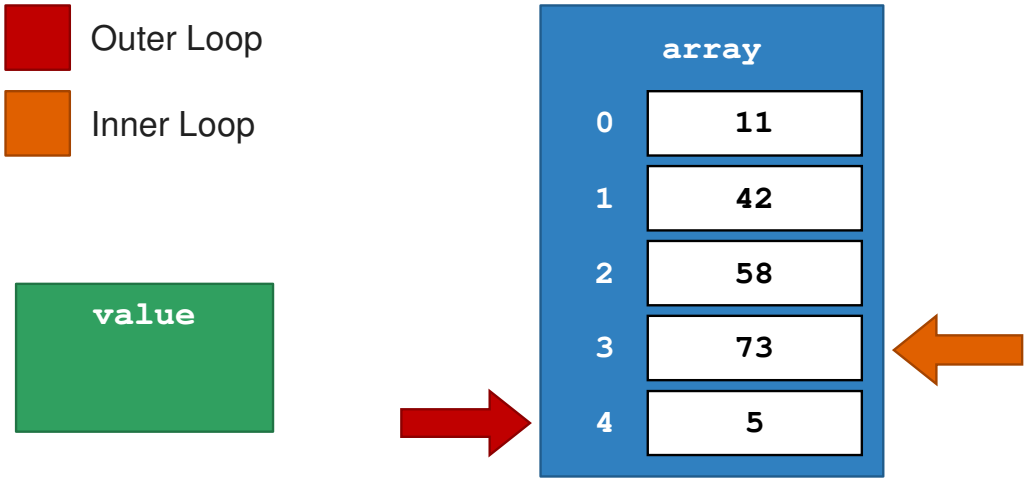
Insertion Sort Example



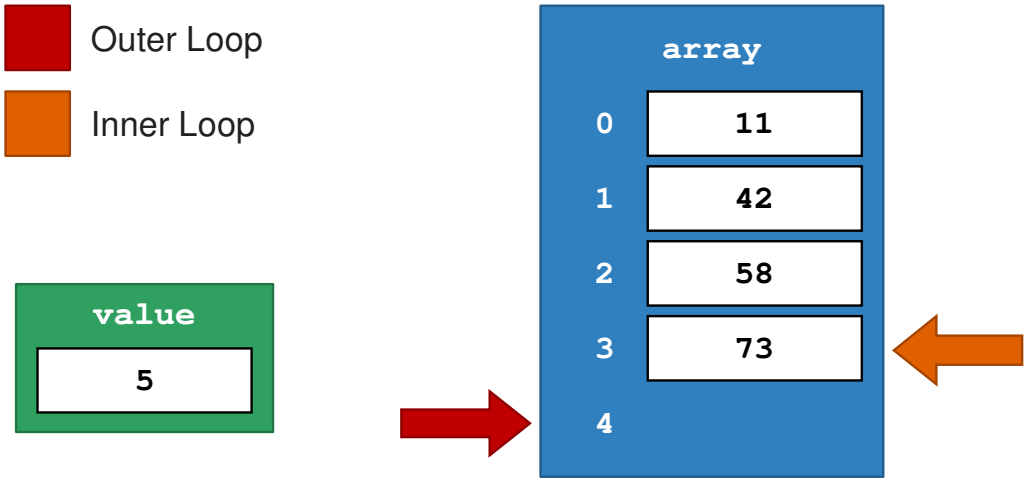
Insertion Sort Example: Return "card"



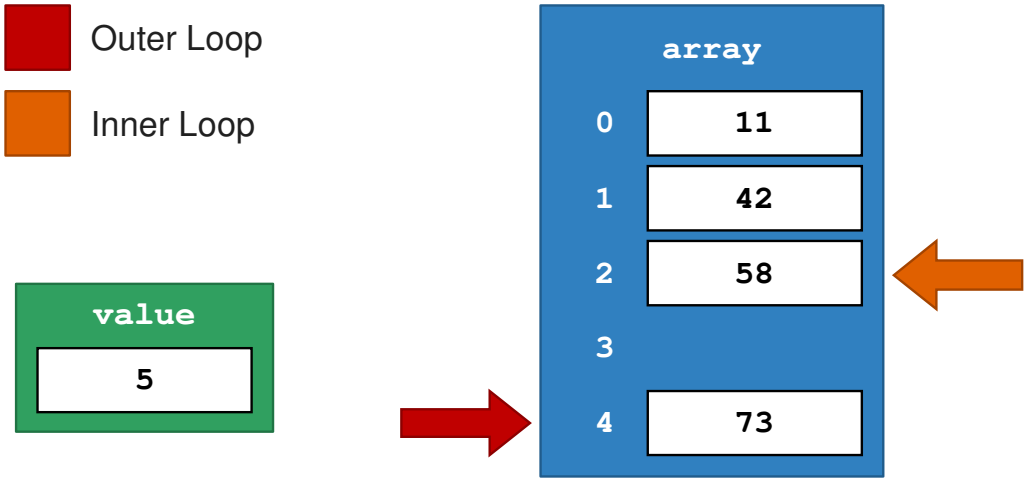
Insertion Sort Example: Next Pass



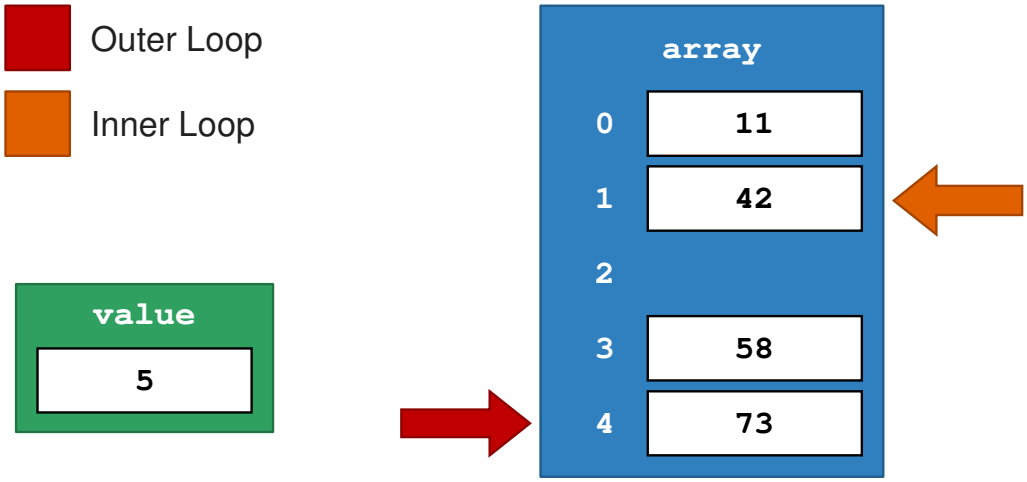
Insertion Sort Example



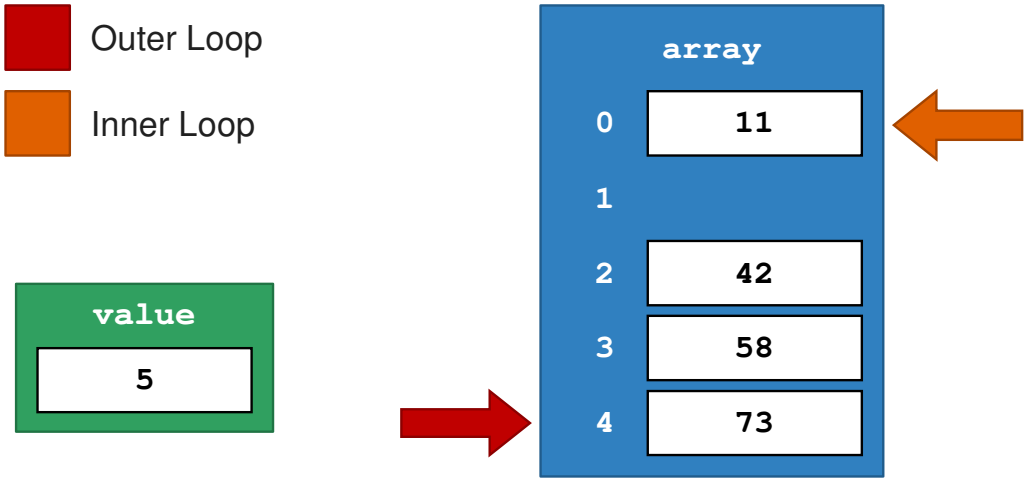
Insertion Sort Example



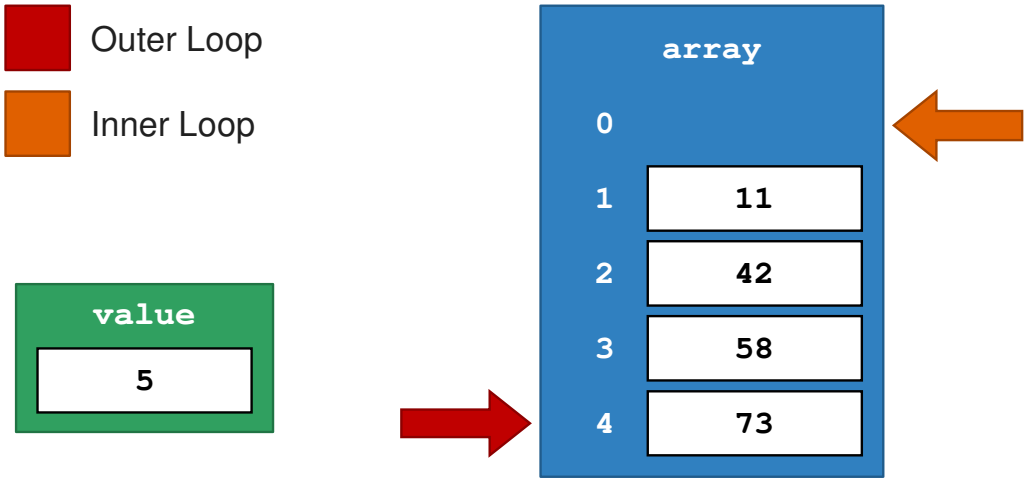
Insertion Sort Example



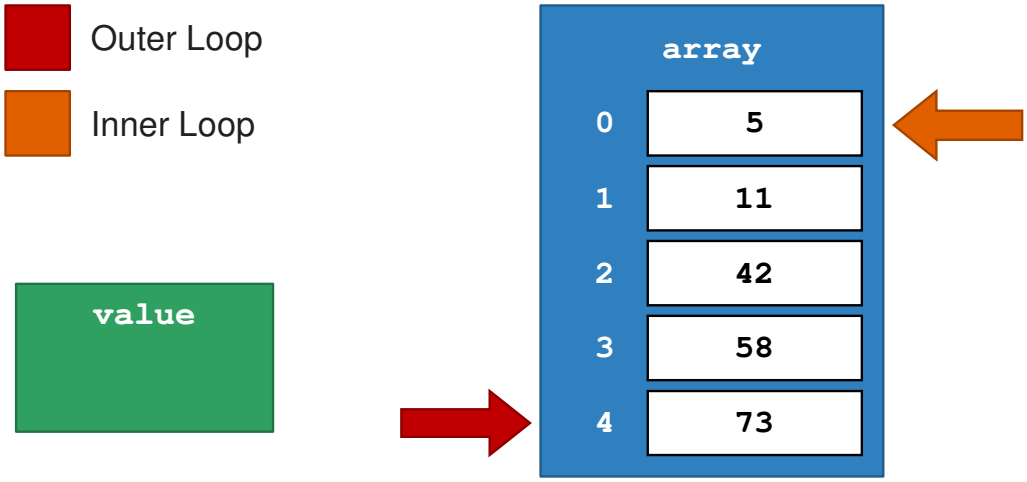
Insertion Sort Example



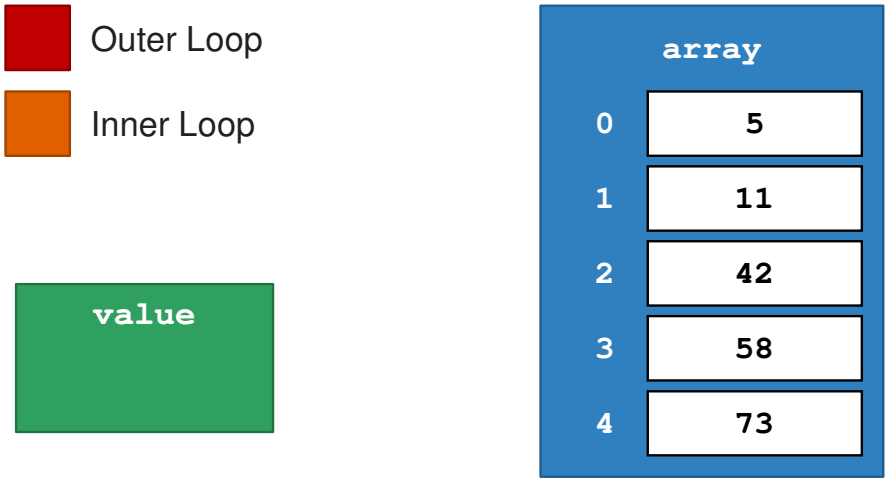
Insertion Sort Example



Insertion Sort Example: Return "Card"



Insertion Sort Example: Done



Advantages

- Because Insertion Sort creates a sorted array above the outer loop
  - inner loop, on average, only needs to move 1/2 positions up – far faster!
  - data can be sent during the sorting process
  - this means the algorithm is considered *"online"* – i.e. it can sort *streaming* data

Advantages

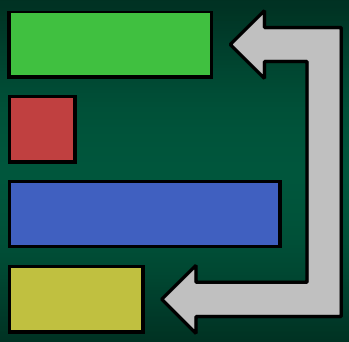
- Insertion sort does not "swap" values
  - most of the overhead of bubble and selection-sort is swapping
  - insertion sort moves data as it sorts, so, there is little unnecessary overhead
- Little to no auxiliary storage overhead
  - like Bubble-Sort and Selection-Sort, Insertion-Sort requires little storage overhead
  - so, in regards to *n*, storage complexity is O(1)

Advantages

- Insertion sort is O(n) on sorted lists
  - inner loop stops when the current array value cannot be moved up
  - the more sorted the list, the more the inner loop approaches O(1)

Insertion Sort Summary

Insertion Sort	
Time Average	$O(n^2)$
Time Best	**O(n)**
Time Worst	$O(n^2)$
Auxiliary space	O(1)
Stable	Yes – Equal element order preserved
Online?	Yes – Can sort streamed data

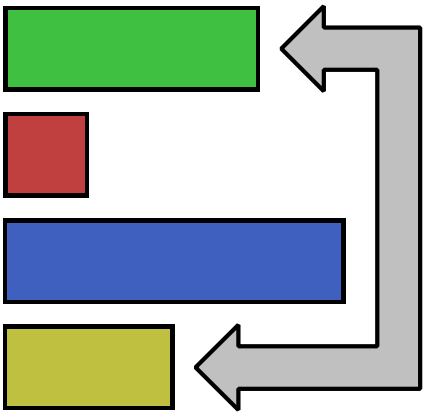


Shell Sort

Insertion Sort with an identity crisis

Shell Sort

- *Shell-Sort* is a version of the Insertion-Sort created by *Donald Shell* in 1959 *(5 BBW)*
- Yes, it is named after the guy, *not a shell metaphor*
- But, ironically, that metaphor works



Shell Sort

- It was the first algorithm to break the $O(n^2)$ barrier
- For a few years, this was the fastest sort algorithm available – until O(n log n) was invented

What is Going On?

- With insertion sort, each time we insert an element, the rest are moved one step closer to where they belong
- Can we move elements a larger distance than just one?
- **Yes**… Shell Sort works like Insertion Sort, but works on elements at large distances
- This distance is called the *gap*

What's Going On?

- Gap changes with each outer loop iteration
  - the distance between comparisons decreases as the sorting algorithm runs
  - in the last iteration, the gap is 1
  - so, at that point, adjacent elements are compared – so it is a regular Insertion Sort
- Shell Sort is also known as a "diminishing increment sort"

Sorting "Shells"

- Shell Sort orders elements that are spaced a relative distance from each other
- So, the red cells above are sorted relative to each other, as are the yellow, green, and blue elements

46	50	92	83	11	36	47	76	53	49	22	3

Sorting "Shells"

- The decreasing gaps are a sequence
- The notation $h_1, h_2, h_3, \ldots, h_t$ represents a sequence of <u>increasing</u> integer values which will be used (from right to left)
- Any sequence works if it $h_n > h_{n-1}$ and $h_1 = 1$

46	50	92	83	11	36	47	76	53	49	22	3

Each Shell is Sorted

- *h_k-sorted array* - all elements with gap h_k are sorted relative to each other
- For each i, we have array[i] ≤ array[i + h_k]
- All elements spaced h_k apart will be sorted

46	50	92	83	11	36	47	76	53	49	22	3

Each Shell is Sorted

- Shell-Sort only works because an array that is h_k-sorted…
- …remains h_k-sorted when h_{k-1}-sorted.

11	36	22	3	46	49	47	76	53	50	92	83

So, What are Gap Values?

- For $h_1, h_2, h_3, \ldots, h_t$ we need to determine what the actual values will be
- Some sequences as better than others
- Shell's original design...
  - starts at N / 2 (where N is the size of the array)
  - cuts the gap in half for each iteration
- There are several competing sequences

So, What are the Gap Values

- The algorithm is most efficient when...
  - the gap sequence are *relatively prime*
  - i.e. the sequence does not share any divisors
- However…
  - using a prime sequence is often not practical in a program – too much to store!
  - so, real, practical solutions attempt to approximate a relatively prime sequence

So, What are Gap Values?

Creator	Sequence
Shell	1, ..., (n / 8), (n / 4), (n / 2)
Hibbard	1, 3, 7, 15, 31, ..., 2^k - 1
Knuth	1, 4, 13, 40, 121, ..., (3^k - 1) / 2
Sedgewick	1, 5, 19, 41, 109, ..., (4^k - 3 * 2^k + 1)

The Shell Sort: Original

```
for (gap = count / 2; gap > 0; gap /= 2)

    for (i = gap; i < array.count; i++)
        value = array[i];
        j = i;

        while j >= gap and  array[j - gap] > value
            array[j] =  array[j - gap]
            j -= gap;
        end while

        a[j + gap] = value
    end for

end for
```

The Shell Sort: Original

```
for (gap = count / 2; gap > 0; gap /= 2)

    for (i = gap; i < array.count; i++)
        value = array[i];
        j = i;

        while j >= gap and  array[j - gap] > value
            array[j] =  array[j - gap]
            j -= gap;
        end while

        a[j + gap] = value
    end for

end for
```

Note that this is an insertion Sort.

All the 1's are replaced with gap.

Gap = 4, First Outer Loop Pass

Outer Loop

Inner Loop

gap

4

value

array

0	73
1	42
2	11
3	58
4	5
5	21
6	7
7	90

Gap = 4, So, 4 overlapped arrays

Outer Loop

Inner Loop

gap

4

value

array

0	73
1	42
2	11
3	58
4	5
5	21
6	7
7	90

Gap = 4, Outer Loop starts at 4 (the gap)

Outer Loop

Inner Loop

gap

4

value

array

0	73
1	42
2	11
3	58
4	5
5	21
6	7
7	90

Inner Loop starts at "gap" indexes up

Outer Loop

Inner Loop

gap

4

value

array

0	73
1	42
2	11
3	58
4	5
5	21
6	7
7	90

Outer Loop (Index 4): Remove 5

Outer Loop

Inner Loop

gap

4

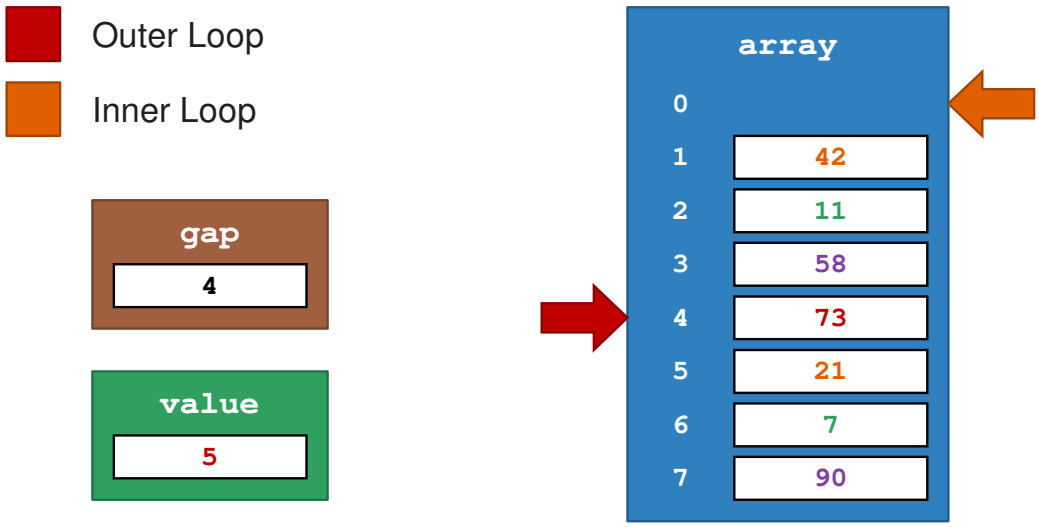
value

5

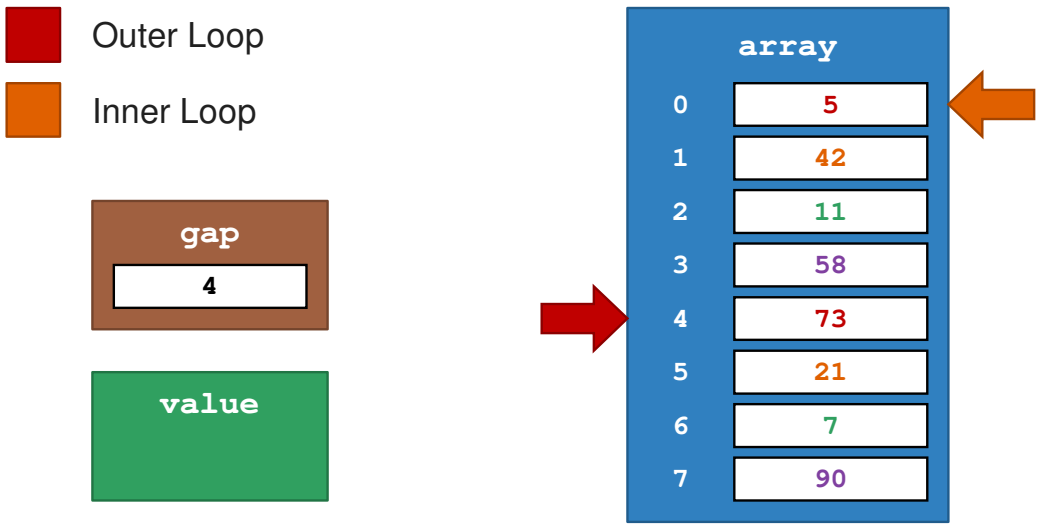
array

0	73
1	42
2	11
3	58
4	
5	21
6	7
7	90

Move value down (Insertion Sort-like)



Inner Loop Done – Return Value



Outer Loop (Index 5): remove 21

Outer Loop

Inner Loop

gap

4

value

array

0	5
1	42
2	11
3	58
4	73
5	21
6	7
7	90

Move 42 Down (its greater than 21)

Outer Loop

Inner Loop

gap

4

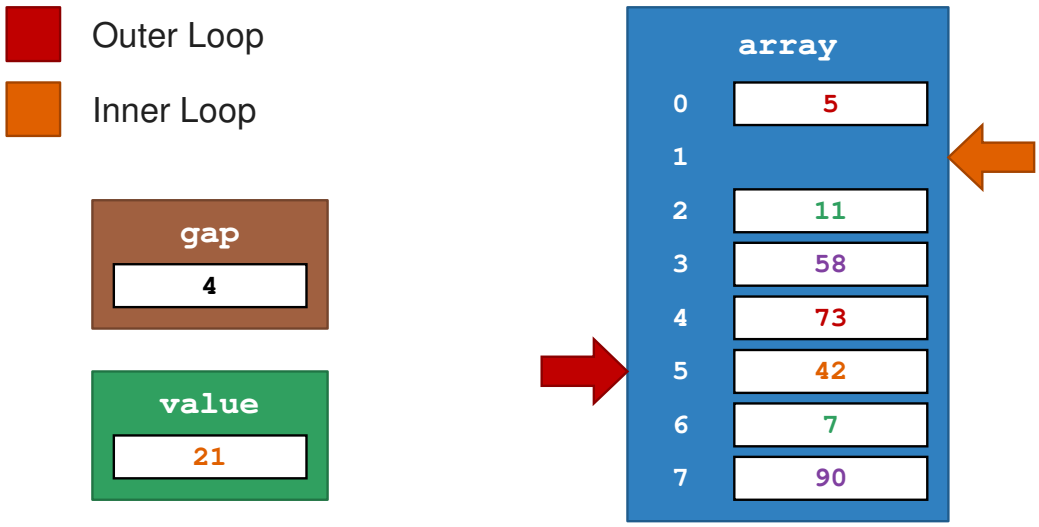
value

21

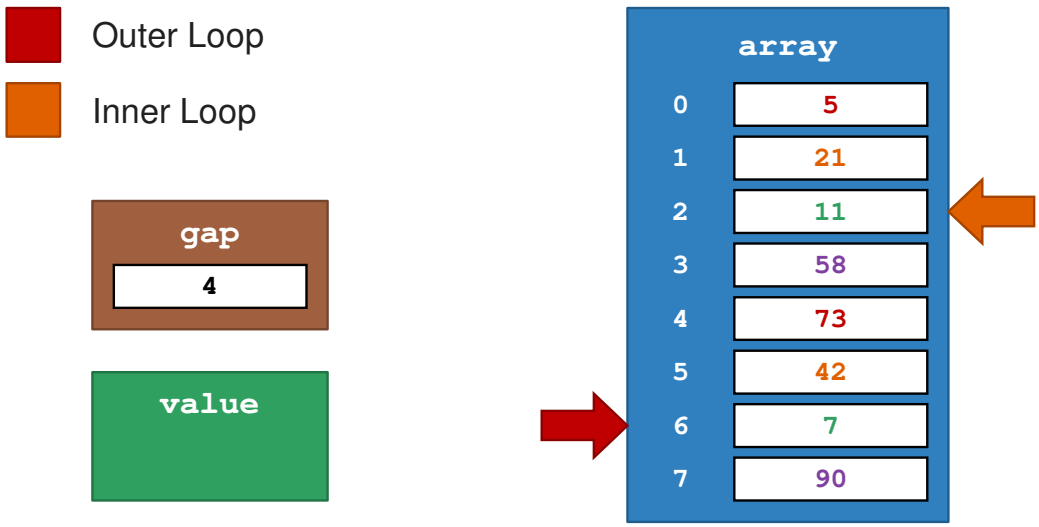
array

0	5
1	42
2	11
3	58
4	73
5	
6	7
7	90

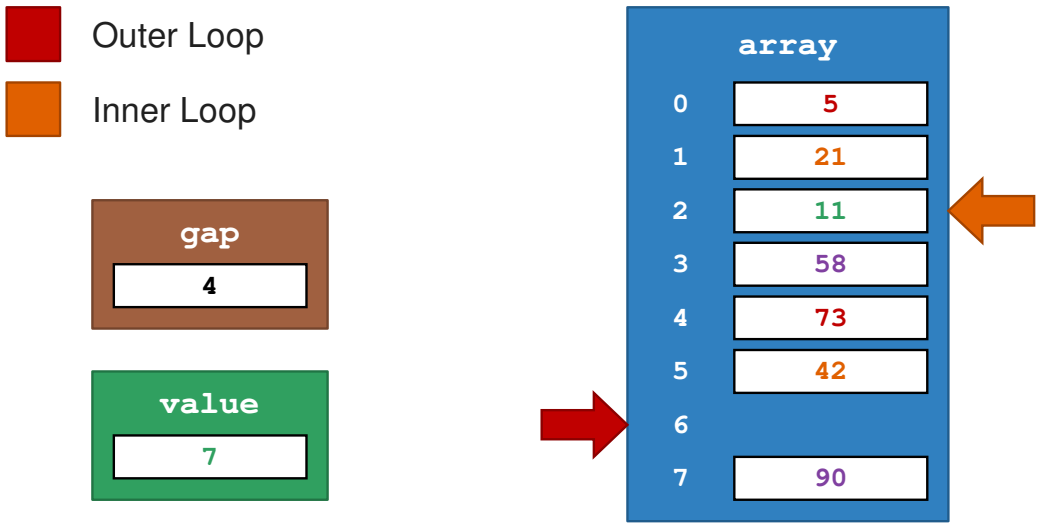
Inner Loop Complete. Return 21



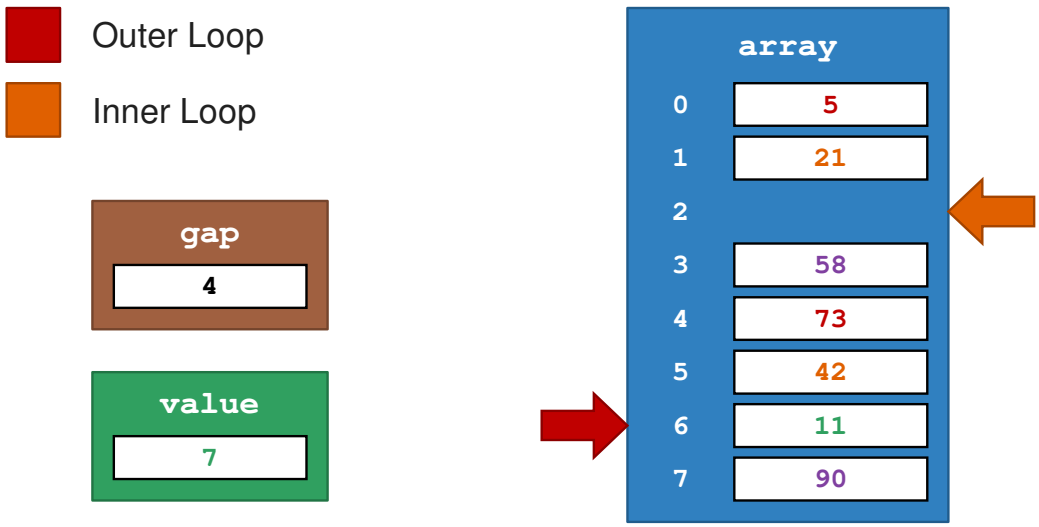
Outer Loop (Index 6): Remove 7



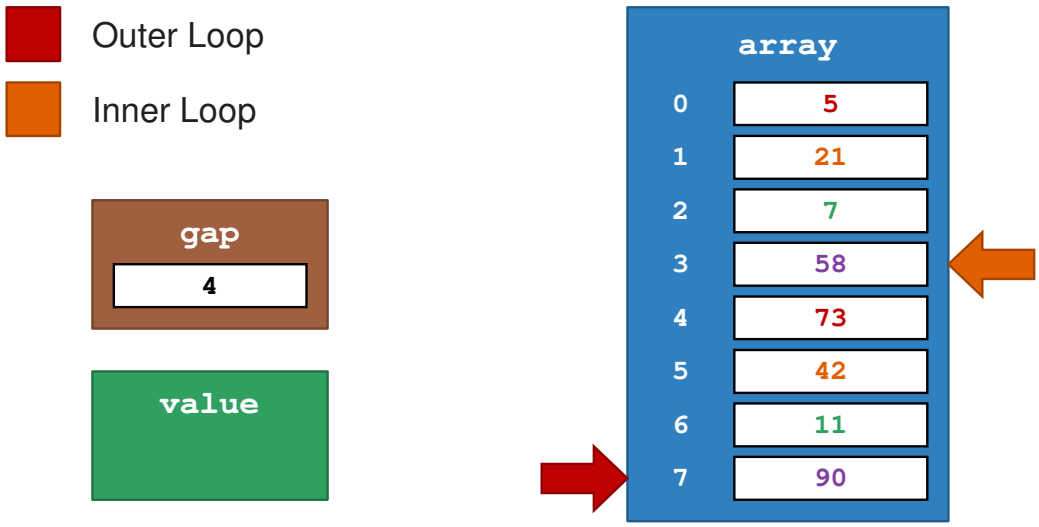
Move 11 Down (its greater than 7)



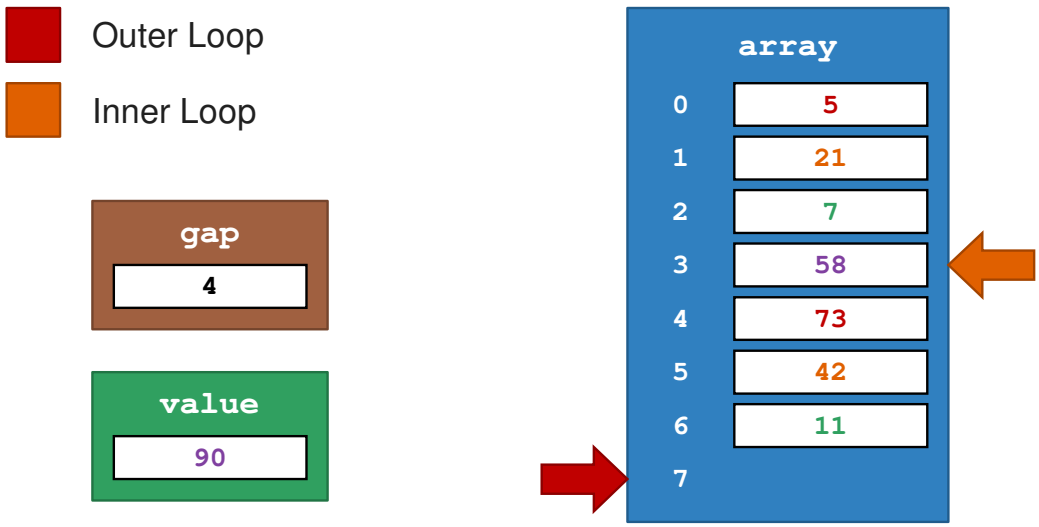
Inner Loop Complete: Return 7



Outer Loop (Index 7): Remove 90



Don't Move 58 Down. It's less than 90.



Inner Loop Complete: Return 90

Outer Loop

Inner Loop

gap

4

value

90

array

0	5
1	21
2	7
3	58
4	73
5	42
6	11
7	

Gap of 4 is Compete

Outer Loop

Inner Loop

gap

4

value

array

0	5
1	21
2	7
3	58
4	73
5	42
6	11
7	90

Note that each overlapped array is sorted

Gap = 2. So now, 2 overlapped arrays

Outer Loop

Inner Loop

gap

2

value

array

0	5
1	21
2	7
3	58
4	73
5	42
6	11
7	90

Gap = 2, Outer Loop #1: Remove 7

Outer Loop

Inner Loop

gap

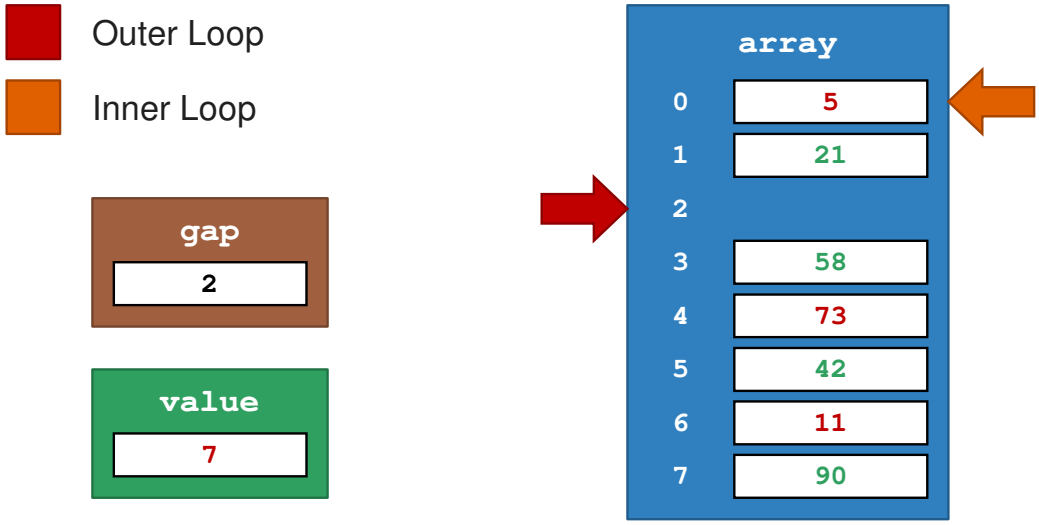
2

value

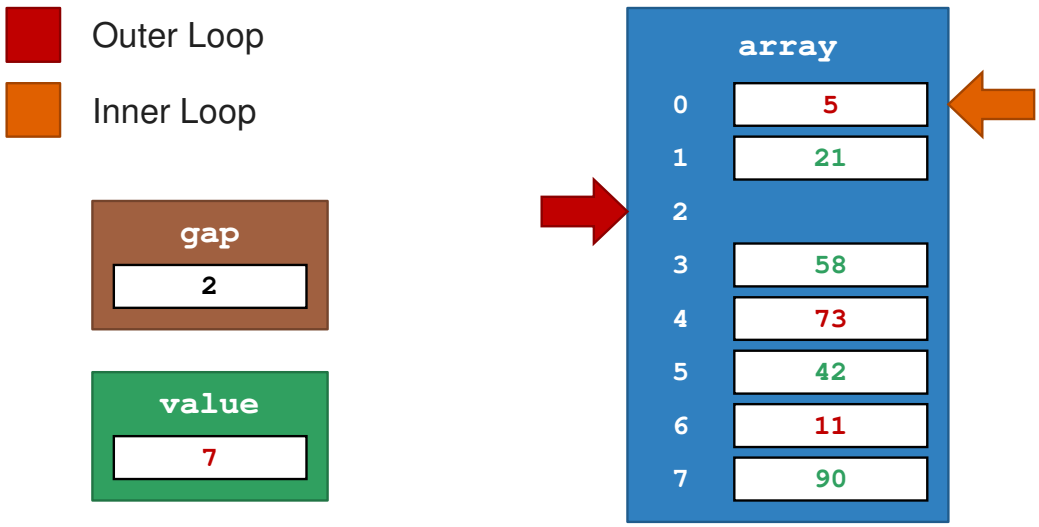
array

0	5
1	21
2	7
3	58
4	73
5	42
6	11
7	90

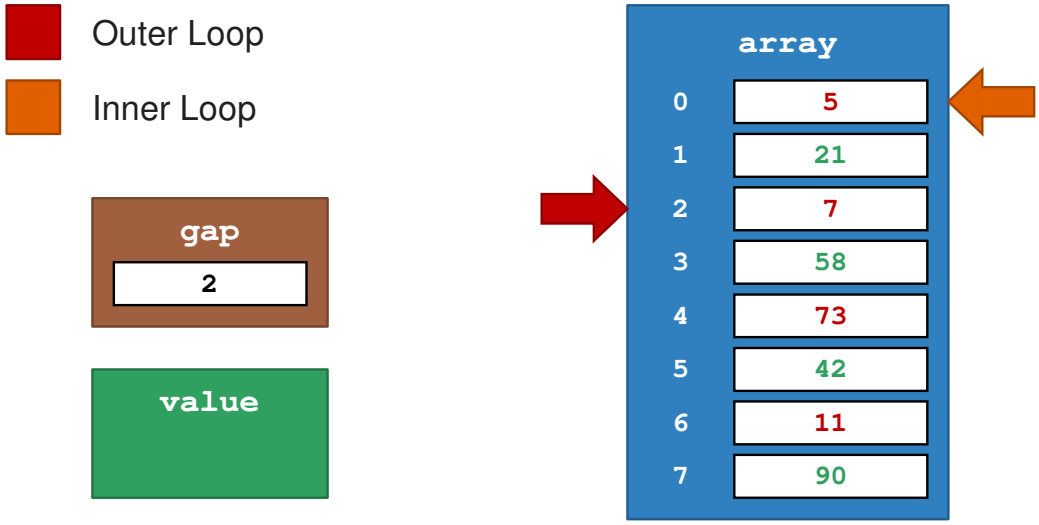
Don't Move 5 Down (smaller than 7)



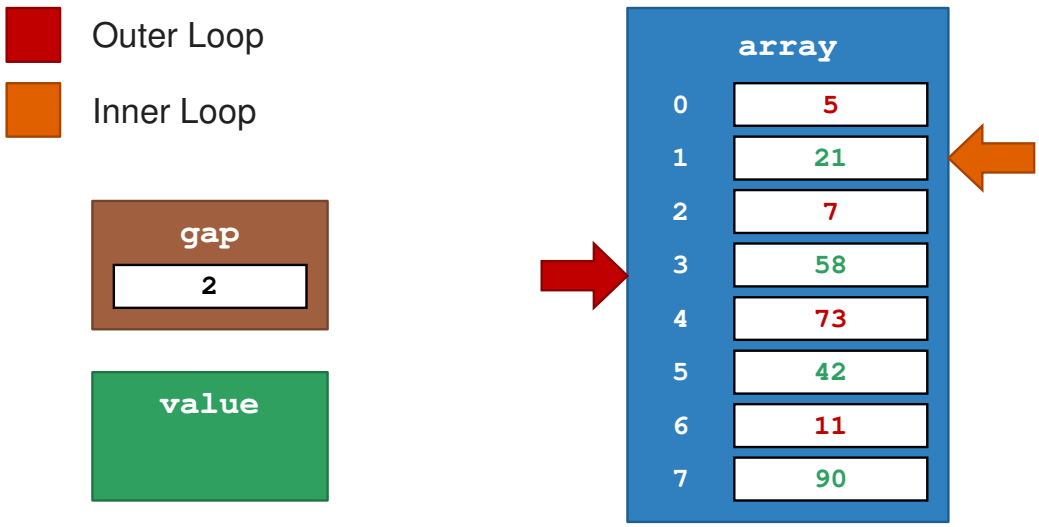
Inner Loop Complete: Return 7



Inner Loop Complete: Return 7.



Outer Loop (Index 3): Remove 58



Don't move 21 down (smaller than 58)

Outer Loop

Inner Loop

gap

2

value

58

array

0	5
1	21
2	7
3	
4	73
5	42
6	11
7	90

Outer Loop (Index 4): Remove 73

Outer Loop

Inner Loop

gap

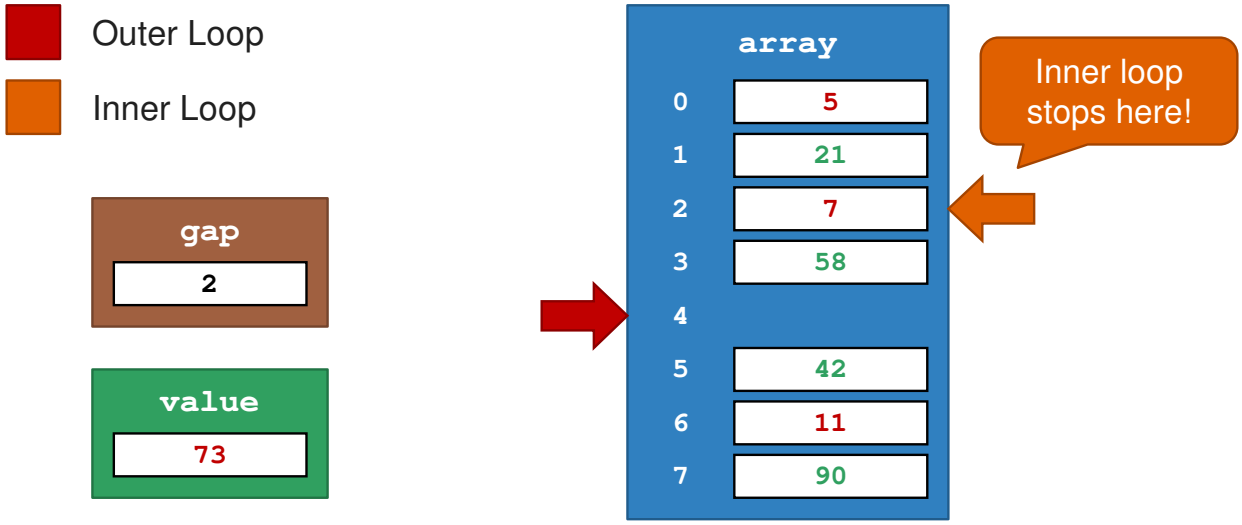
2

value

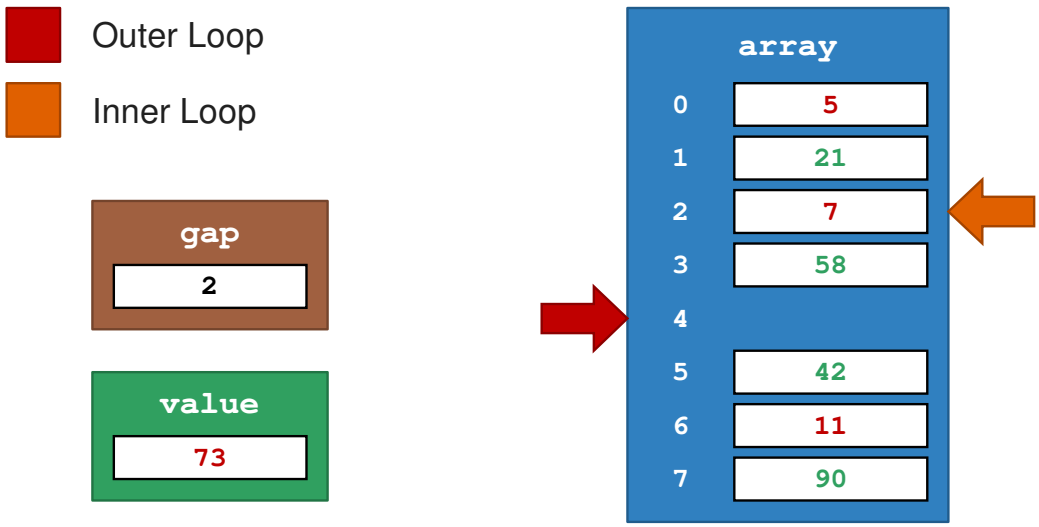
array

0	5
1	21
2	7
3	58
4	73
5	42
6	11
7	90

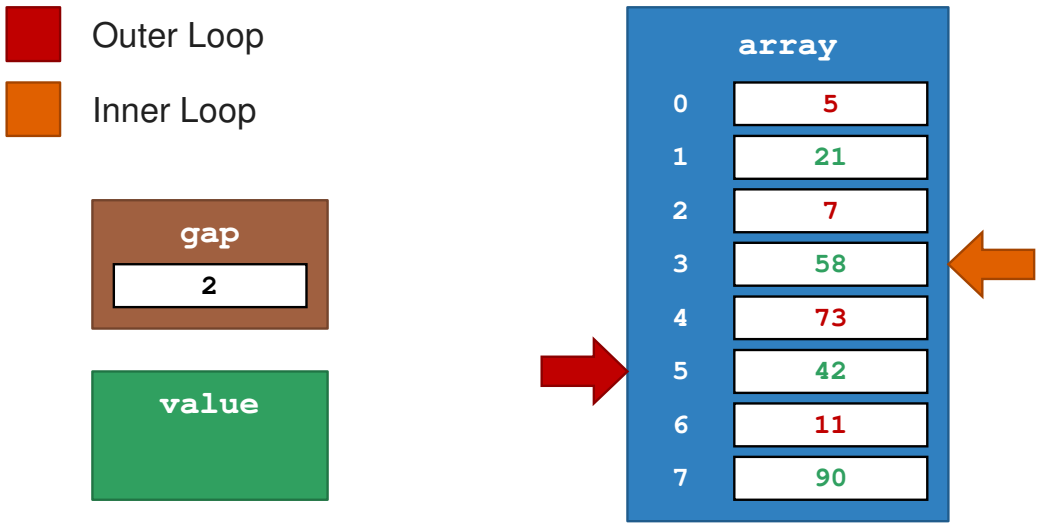
Don't move 7 down. Inner Loop Stops.



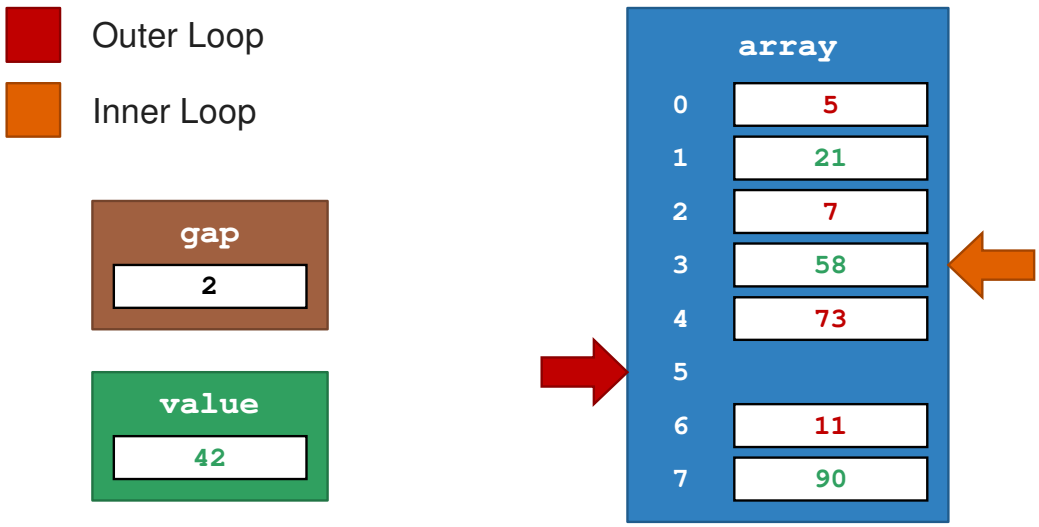
Inner Loop Complete: Return 7.



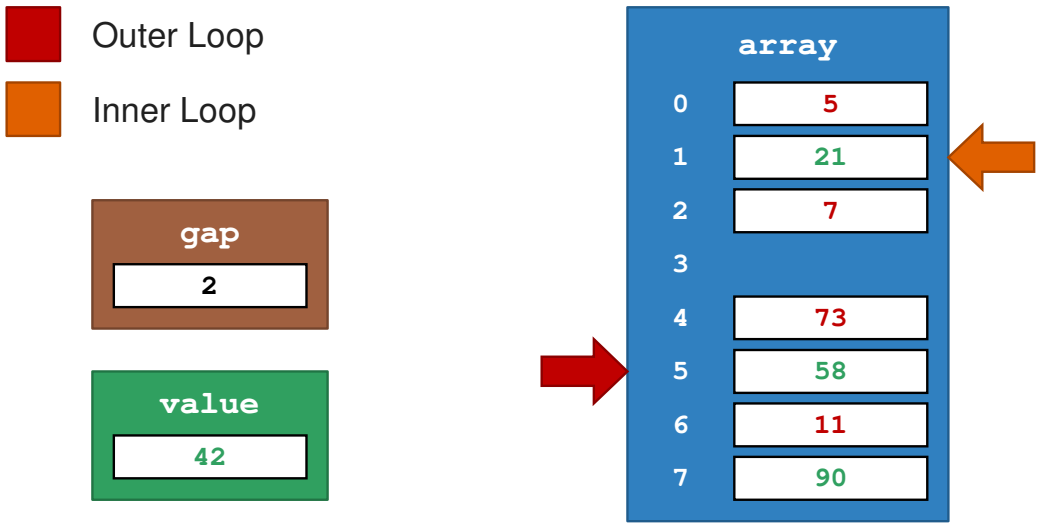
Outer Loop (Index 5): Remove 42



Move 58 down (it's greater than 42)



Inner Loop Complete. Return 42



... and so on....

- The example continues to sort for each h_k
- The outer loop continues to the bottom of the array
- Finally, gap will go to one and the sort acts just like an Insertion-Sort

Gap of 2 is Compete

Outer Loop

Inner Loop

gap

2

value

array

0	5
1	21
2	7
3	42
4	11
5	58
6	73
7	90

Note that each overlapped array is sorted

Big-O is still being determined

- Although the algorithm is fairly simple, proving its time complexity is not
- Time complexity of Shell Sort is **still** up for debate
- *It's an unsolved computer science riddle!*

Big-O is still being determined

- Empirical analysis of the algorithm has given some widely accepted values…
  - it is approximately $O(n^r)$ where $1 < r < 2$
  - this is ultimately faster than $O(n^2)$ but worse than $O(n \log n)$
- Worst case (using Hibbard's sequence) is $O(n^{3/2})$
- Average performance is thought to be about $O(n^{5/4})$

Shell Sort Summary

Shell Sort	
Time Average	≈ $O(n^{5/4})$
Time Best	≈ O(n log n) – For a near sorted list
Time Worst	≈ $O(n^{3/2})$
Auxiliary space	O(1)
Stable	No – Equal element order not preserved
Online?	No – Entire array in use